130. How high altitudes affect our breathing system? At high altitude
A. The number of red blood cells increases, as available oxygen increases.
B. The number of red blood cells decrease as available oxygen decreases.
C. The level of oxygen is low, as a result our breathing rate increases
D. The level of oxygen is high, as our breathing rate decreases.

Explanation: As we go higher and higher from the place of lower altitude to higher altitude, the air becomes less and less denser. Thus, since the level of air including oxygen gets lower, the rate and depth of breathing increases.

Answer: C

131. Assume Mr, Y has a monthly salary of 700 birr and he has three children. He smokes cigarettes and chews khat, frequently. What will be the fate of his family? (EGSECE, 2006 E.C)
A. Mr. Y and his family will be healthier.
B. The family will suffer from shortage of income.
C. Nothing will happen to his family.
D. Mr. Y and his family will live cheerfully.

Explanation: Drug addiction may cause various problems on an individual, family, community, country even the world. Thus, among various problems related to drug addiction, some of them are:

- Economic crisis
- Psychological problem
- Health problem

132. This item is based on the following reaction (EGSECE, 2006 E.C)

$Z + Pi \rightleftharpoons ATP$

Answer: B

In the above reaction "Z" stands for
A. O_2 B. ADP C. CO_2 D. H_2O

Explanation:
ATP (adenosine tri-phosphate) is the direct or immediate source of energy for the cells. It is made from ADP (Adenosine di-phosphate), pi (phosphate) and energy that released from the

breakdown of glucose. Thus, the chemical reaction by which ATP is formed can be indicated as:

$$ADP + pi + energy \rightleftharpoons ATP$$

Answer: B

133. This item is based on the following table which shows the comparison between aerobic and anaerobic respiration.

No.	Aerobic respiration	Anaerobic respiration
1	Glucose is the reactant.	Glucose and oxygen are the reactants.
2	Undertaken in the absence of oxygen	Undertaken in the presence of oxygen
3	Carbon dioxide and water are products	Ethanol and carbon dioxide are products
4	Takes place in mitochondria of animal cells.	Takes place in muscle cells of animals

Which of the following comparisons is true about aerobic and anaerobic respiration in cells? (EGSECE, 2006 E.C)

A. 4 B. 1 C. 1, 2, 4 D. 1,2,3

Explanation:

The correct comparision between aerobic and anaerobic respiration is:

Aerobic respiration	Anaerobic respiration
Glucose and oxygen are the reactants	Glucose is the reactant
Undertake in the presence of oxygen	Under take in the absence of oxygen
Carbon dioxide and water are products	Ethanol and carbondioxide are products
Takes place in the mitochondria of the cell	Takes place in the cytoplasm of the cell and muscle cell during heavy exercise

Answer: 3 and 4

Post Test 6

Part I: Choose the best answer from the given alternatives.

1. The type of blood that is known as a universal donor is:

A. Type 'A' C. Type 'AB'

B. Type 'B' D. Type 'O'

2. A blood type with no antigen is:
   A. Type 'AB' C. Type 'A'
   B. Type 'O' D. Type 'B'
3. Red blood cells are used in:
   A. Clotting blood
   B. Defending disease
   C. Transporting oxygen
   D. Preventing hypertension
4. The types of white blood cells that engulf disease causing microorganisms are:
   A. Lymphocytes C. Phagocytes
   B. Thrombocytes D. Platelets
5. Which of the following blood transfusion may result in agglutination?

	Donor	Recipient
A	Type 'A'	Type 'AB'
B	Type 'O'	Type 'A'
C	Type 'AB'	Type 'AB'
D	Type 'B'	Type 'O'

6. All of the following are solid parts of the blood except:
   A. RBC B. Platelets C. RBC D. Plasma
7. The blood cells that avoid excessive loss of the blood when we are exposed to cut or wound?
   A. WBC B. RBC C. Platelets D. Plasma
8. The shortage of hemoglobin in the RBC may cause:
   A. Hypertension C. Anemia
   B. Leukemia D. Elephantiasis
9. Hypertension cannot be reduced by:
   A. Regular exercise C. Healthy life style
   B. Eating more salt D. Eating less fat

Supplementary Exercises

1. Of the following groups of nutrients which one is required by the body in a very small amount?
   A. Carbohydrate C. proteins
   B. Vitamins D. lipid
2. Among the following types of carbohydrates, which one is a complex sugar?
   A. Fructose C. cellulose
   B. Galactose D. Maltose

3. The sugar which is found in milk is called:
   A. Sucrose C. Maltose
   B. Lactose D. Galactose
4. All of the following are correct about carbohydrate, except.
   A. It's a macronutrient.
   B. It's the main source of energy for the body
   C. It's the storage form of energy in the body
   D. It helps to build and repair the tissues.
5. Which of the following is correct:
   A. Glucose + fructose $\longrightarrow$ lactose
   B. Galactose + glucose $\longrightarrow$ maltose
   C. Glucose + fructose $\longrightarrow$ sucrose
   D. Glucose + glucose $\longrightarrow$ lactose
6. An element which is used in making proteins, but not found in carbohydrate and lipids is:
   A. Carbon C. Nitrogen
   B. Oxygen D. Hydrogen
7. The process by which soluble food passes through the walls of the small intestine into the blood vessels is:
   A. Egestion B. Ingestion C. digestion D. absorption
8. One of the following statements is correct.
   A. All enzymes work best in acidic pH.
   B. Most human enzymes work best at body temperature
   C. Enzymes are proteins that undergo chemical change
   D. They are destroyed in the process of the reaction.
9. Enzymes that perform their activities outside the cell are called
   A. Respiratory enzymes C. extra cellular enzymes
   B. Intracellular enzymes D. biological enzymes
10. Human digestive enzyme that optimally function at pH around neutrality is:
    A. Pepsin B. Renin C. ptylin D. Typsin
11. A person whose gallbladder is surgically removed is be careful to restrict his dietary intake of
    A. Starch B. protein C. water D. fat
12. Which pair is not correctly matched?
    A. Glucose – monosaccharide
    B. Galactose – disaccharide
    C. Cellulose – polysaccharide
    D. Sucrose - disaccharide

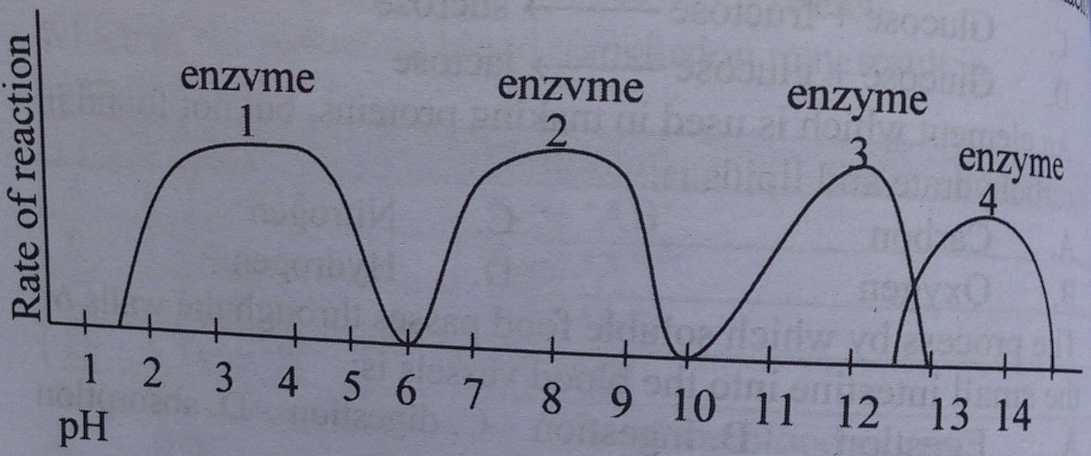
13. Which of the following alternative contains the correct mate... nutrient, name of test and color observed when the test result... positive?

A. Protein – Fehling test → blue-black color
B. Starch – Buriet test → purple color
C. Fat – oil spot test→translucent spot
D. Glucose – iodine test → Brick red color

14. Which nutrient does the human stomach principally digest?

A. Carbohydrates
B. Fats
C. proteins
D. vitamins

15. Which enzyme is likely to be most active in the human stomach?



A. Enzyme 1
B. Enzyme 2
C. enzyme 3
D. enzyme 4

16. Rickets is caused by the lack of

A. Iron and calcium
B. Iron and phosphorus
C. Calcium and phosphorous
D. Iodine and phosphorus

17. Pepsin is a digestive enzyme that acts on:

A. Proteins
B. Carbohydrates
C. fats
D. vitamins

18. The pancreatic juice does not contain one of the following enzymes.

A. Trypsin
B. Lipase
C. pancreatic amylase
D. Lactase

19. The enzyme ptylin functions best at a

A. Slightly acidic medium
B. Slight basic medium
C. Neutral medium
D. strongly acidic medium

20. Which one of the following is NOT true about proteins?
A. They are made up of amino acids.
B. Buriet reagent turns proteins blue
C. They form the main tissues of organisms
D. Some proteins act as catalysts.

21. Chemical digestion begins in the :
A. Mouth
B. Esophagus
C. stomach
D. small intestine

22. The reason why there are so many different kinds of enzymes in a cell is that,
A. An enzyme can be used over and over again
B. Enzymes are very specific
C. The activity of enzymes can be affected by high temperature
D. The activity of enzymes can be affect by low pH.

23. If equal amounts of the following substances are metabolized by the body, the largest amount of energy is released from
A. Carbohydrate
B. Protein
C. Fat
D. vitamin

24. Choose the incorrectly matched enzyme with its secreting gland.
A. Ptylin – salivary glands
B. Lipase – intestinal glands
C. pepsin – gastric glands
D. Erepsin – intestinal glands

25. Which of the following alternatives consists of the end products of digestion.
A. Sugar, proteins, fat and glycerol
B. Amino acids, glucose, fatty acid and glycerol
C. Polysaccharides, proteins, fats and oils
D. Glucose, triglycerides, maltose, glycoproteins

26. Which of the following cannot be directly absorbed into the villi which are the walls of the small intestine?
A. Proteins
B. Fatty acids
C. glucose
D. amino acids

27. Which of the following reactions take place when a muscle is contracting?
A. ADP + phosphate + energy ⟶ ATP
B. ATP ⟶ ADP + phosphate + energy
C. ATP + phosphate ⟶ ATP + energy
D. ATP + phosphate ⟶ ADP + energy

28. Which of the following valves of the heart is opened by the blood forced out of the right ventricle?
A. Tricuspid valve
B. Bicuspid valve
C. semilunar valves of the aorta
D. semilunar valves of the pulmonary artery

29. The windpipes such as trachea and bronchi are always kept open by means of
A. Arterial blood pressure
B. Atmospheric gas pressure
C. muscles in their walls
D. rings of cartilages

30. The chamber of the heart which pumps blood over a longer distance than the rest?
A. Left auricle
B. Right auricle
C. left ventricle
D. right ventricle

31. What does the red blood cell lack that the white blood cell has?
A. Hemoglobin
B. Nucleus
C. cell membrane
D. cytoplasm

32. The correct order of air passage during inhalation, takes places through:
A. Bronchus ⟶ trachea ⟶ bronchiole ⟶ alveolus
B. Trachea ⟶ bronchus ⟶ bronchiole ⟶ alveolus
C. Trachea ⟶ bronchiole ⟶ bronchus ⟶ alveolus
D. Bronchus ⟶ bronchiole ⟶ trachea ⟶ alveolus

33. Which of the following is true about anaerobic respiration?
A. ATP is produced
B. Lactic acid is degenerated
C. Glucose releases all its energy
D. Oxygen is required

34. The quickest source of energy for the muscle is:
A. Glucose sugar
B. Maltose sugar
C. starch
D. ATP

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37. Which
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38. Which
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40. If
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Items N
parts.

41.

42.

35. The exchange of materials between the blood and tissues of the body takes place in the
A. Venuole B. Arteries C. arterioles D. capillaries

36. In the process of food oxidation in a cell, the energy released from the food is trapped by molecules of
A. Glucose C. adenosine triphosphate
B. oxygen D. adenosine diphosphate

37. Which of the following has the thickest wall?
A. Left auricle C. Right auricle
B. Left ventricle D, right ventricle

38. Which of the following blood diseases are heritable?
A. Leukemia C. leukemia
B. Hemophilia D. hypertension

39. Which of the following is correct sequence of blood flow in mammals?
A. Pulmonary vein → left auricle→ left ventricle → Aorta
B. Pulmonary vein ⟶ right auricle⟶ pulmonary artery⟶lung
C. Vena cava ⟶ right auricle ⟶ pulmonary vein ⟶ left ventricle
D. Left auricle ⟶ left ventricle ⟶ pulmonary artery ⟶ lung

40. If you cut your finger, blood comes out and then stops flowing, this is due to clotting of blood by.
A. platelets C. white blood cells
B. Antibodies D. Red blood cells

Items No 41 is based on the following structures of human circulatory parts.

1. Right atrium
2. Right ventricle
3. Left atrium
4. Left ventricle
5. Pulmonary artery
6. Bicuspid valve
7. Tricuspid valve
8. Aorta
9. Pulmonary vein

41. Which of the following are sequentially arranged
A. 1,2,7 B. 1,2,6 C. 3,4,8 D. 2,8,9

42. If a person climbing a high mountain was found to be taking faster and deeper breaths. This shows that.
A. The proportion of oxygen in the air he is breathing is much more lower than in the lower altitude.

B. At higher altitude the rate of breathing is much lower
C. Air at high altitude is less dense and a person would need to breathe more quantity of air.
D. The percentage of oxygen in the air decreases.

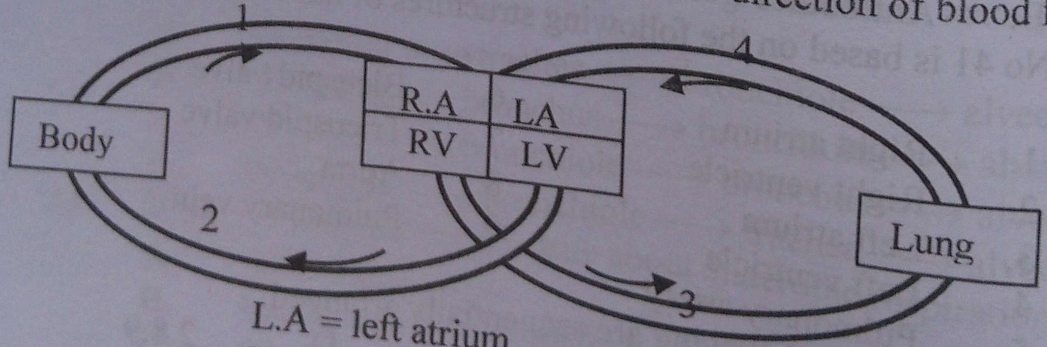
43. Which of the following statement is true?
A. We breathe in oxygen and breath out carbondioxide.
B. The air we breathe out contains more carbondioxide than the air we breathe in.
C. The air we breathe in and we breath out contain the same amount of oxygen.
D. The exhaled air contains more carbondioxie than oxygen

44. During the process of breathing in air
A. The lung inflates
B. The volume of chest cavity decreases
C. The diaphragm becomes dome shaped
D. The rib muscles relax

45. The structure that protects the windpipe from being choked with food is the
A. Esophagus
B. Epiglottis
C. rings of tracheal cartilage
D. larynx

46. Which of the blood vessels has the greatest surface area?
A. Arteries
B. Veins
C. capillaries
D. Arterioles

Questions 47 to 49 are based on the following simplified diagram of the human circulatory system. Arrows indicate the direction of blood flows.



L.A = left atrium
L.V = Left ventricle
R.A right atrium
R.V right ventricle

47. The blood vessel number 1 is
A. Pulmonary vein
B. Venacava
C. pulmonary artery
D. Aorta

48. The dia
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59.

48. The diagram represents
A. Open circulation
B. Double circulation
C. aortic circulation
D. simple circulation

49. Which of the following blood vessels carry deoxygenated blood
A. 1 and 3
B. 1 and 2
C. 1 and 4
D. 3 and 2

50. The valve which is found between right atrium and right ventricles is:
A. Bicuspid valve
B. Tricuspid valve
C. mitral valve
D. semilunar valve

51. Which two chambers of the human heart contain oxygenated blood?
A. Right auricle and left ventricle
B. Left ventricle and left auricle
C. Left auricle and right ventricle
D. Right ventricle and right auricle

52. Which of the following explains sequential steps of nutrition?
A. Absorption→ digestion → Ingestion → assimilation
B. Digestion → Ingestion → Assimilation→ Absorption
C. Ingestion → Digestion → Absorption→ Assimilation
D. Ingestion → Digestion → Assimilation→ Absorption

53. Which of the following statement is **incorrect** about bile?
A. It helps to emulsify fat
B. It is secreted by the liver
C. It has enzymatic value
D. It is stored in gall bladder

Fill in the black spaces.

54. _____ is the process of obtaining and utilizing food.
55. The churning movement of the food along the oespohagus is called _____.
56. _____ is the right type and precise amount of food required for the healthy state of the body.
57. Carbohydrates are made up of elements _____, _____ and _____.
58. _____ are biological catalysts.
59. A fat emulsifying salt which is produced by the liver is called _____.

60. The deficiency disease of vitamin – A is called _____.
61. Simple sugars join to from a double sugar by the reaction known as ___
62. The storage form of carbohydrates in plants is _____, while the storage form in animals is called _____.
63. The part of the human digestive system which has both digestive and absorption role is _____,
64. The chemical digestion of carbohydrate starts in the _____ and ends in the _____.
65. The three enzymes that are found in pancreatic juice are _____, _____ and _____.
66. The process of taking air in to the lung is called _____, while the act of expelling air from the lung is called _____.
67. Involuntary muscles of the heart are called _____.
68. The entrance food and dust particles in to the wind pipe is prevented by_____.
69. The functional units of the lung where exchange of gas takes place are called _____.
70. _____ is the amount of oxygen needed to break the lactic acid produced during the anaerobic respiration in animals.
71. The powerful addictive chemical which is found cigarette smoke is called _____.
72. The largest veins are called _____, while the smallest veins are called _____.
73. The blood vessels that carry the blood away from the heart are _____.
74. The contraction of the muscles of the heart is _____, while the relaxation of the muscles of the heart is _____.
75. The back flow of the blood in the human circulatory system is prevented by _____.
76. Blood type 'O' is called _____, while blood group 'AB' is called ____.
77. The uncontrolled production of white blood cells leads to _____
78. The disease which is characterized by failure of the body to clot blood after injury is _____.

Part I: **Answer to Post Test 1**

1. D 2. B 3. B 4. A 5. C
6. B 7. B 8. C 9. C 10. A

Part II: 1. I 2. C 3. J 4. G 5. A
6. F 7. H 8. D 9. E 10. B

Answers to Post Test 2

1. C 2. A 3. C 4. B 5. B
6. D 7. B 8. B 9. C 10. A

Answer to Post Test 3

1. C 2. A 3. C 4. C 5. A 6. B 7. B 8. D

Answers to Post Test 4

1. D 2. B 3. C 4. B 5. D

Answers to Post Test 5

Part I: 1. D 2. B 3. B 4. C 5. B
6. C 7. A 8. A 9. B 10. C 11. B

Answers to Post Test 6

1. D 2. B 3. C 4. C 5. D 6. D 7. C 8. C 9. B

Answers to Supplementary Exercises

1. B 19. C 37. B 54. Nutrition
2. C 20. B 38. B 55. Peristalsis
3. B 21. A 39. A 56. Balanced diet
4. D 22. B 40. A 57. Carbon, hydrogen, oxygen
5. C 23. C 41. C 58. Enzymes
6. C 24. B 42. C
7. D 25. B 43. B 59. Bile
8. B 26. B 44. A 60. Night blindness
9. C 27. B 45. C 61. Condensation reaction
10. C 28. D 46. C 62. Starch, glycogen
11. D 29. D 47. B 63. Ileum
12. B 30. C 48. B 64. Mouth, Ileum
13. C 31. B 49. A 65. Trypsin, amylase, lipase
14. C 32. B 50. B 66. Inhalation, Exhalation
15. B 33. A 51. B 67. Cardiac muscle
16. C 34. D 52. C 68. Epiglotis
17. A 35. D 53. C 69. Alveoli
18. D 36. C 70. Oxygen dept
71. Nicotine
72. Venacava, Venuoles
73. Arteries
74. Diastole, systole
75. Valves
76. Universal donor, Universal recipient
77. Leukemia
78. Hemophilia

4. Microorganisms and Disease

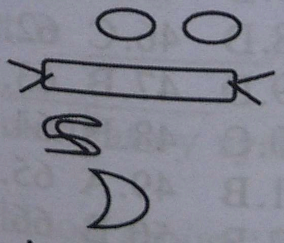
4.1. Microorganisms

Microorganisms are organisms which are too small to be seen by naked eye. They can be seen with the aid of the microscope.
Groups of organisms which are classified as microorganisms include; bacteria's, viruses, protozoan, molds, yeast, etc…

1. Bacteria

- are unicellular and prokaryotic organisms.
- have cell wall, cell membrane, cytoplasm and genetic materials.
- Some bacteria have additional features like **flagella** (for movement) and **capsule** (for protection).
- have varied mode of nutrition. Some are autotrophic while others are heterotrophic.
- Most of them are harmless and useful. While some of them are harmful, since they produce toxic substances in to the body and cause diseases.
- Bacteria are classified based on their shapes.

Types of bacteria	Shape
Cocci	spherical
Bacilli	Rod
Sprilli	spiral
Vibro	comma



The useful aspects of bacteria include:

- ✓ **Making of cheese and yoghurt**. It's carried by the action of bacteria's on the milk sugar (lactose). The action of bacteria on lactose forms **lactic acid**. The lactic acid makes a milk to clot to form cheese and yoghurt.
- ✓ Making of vaccines and antibiotics
- ✓ Tanning of lather in industries
- ✓ Increase the soil fertility by their decomposing effects

2. Viruses

Viruses are considered to be intermediates between living and non-living things. This is because viruses act as a living thing and reproduce only inside the host cell. They

- are **obligate parasites**
- have either DNA or RNA as genetic material
- cause some human diseases

Examples are HIV AIDS, influenza, rabies

3. Fungi

Funguses are heterotrophic organisms that have the following features in common. They:

- ✓ are decomposers, free living or parasitic;
- ✓ are either multicellular or unicellular;
- ✓ are useful, harmful or harmless;

Examples: yeast,

i. Yeast

Yeast is a single celled fungus which plays a great role in facilitating our life.

At anaerobic conditions yeast carryout **alcoholic fermentation**

$$C_6H_{12}O_6 \xrightarrow[\text{Fermentation}]{\text{Yeast}} 2C_2H_5OH + CO_2 + 2ATP$$

(Ethyl alcohol)

Thus, they play role in preparation of alcoholic drinks like tej, tella, beer etc.

Yeasts

- ✓ have nucleus, cytoplasm, cell membrane and cell wall.
- ✓ have heterotrophic mode of nutrition like all fungi
- ✓ reproduce asexually by budding

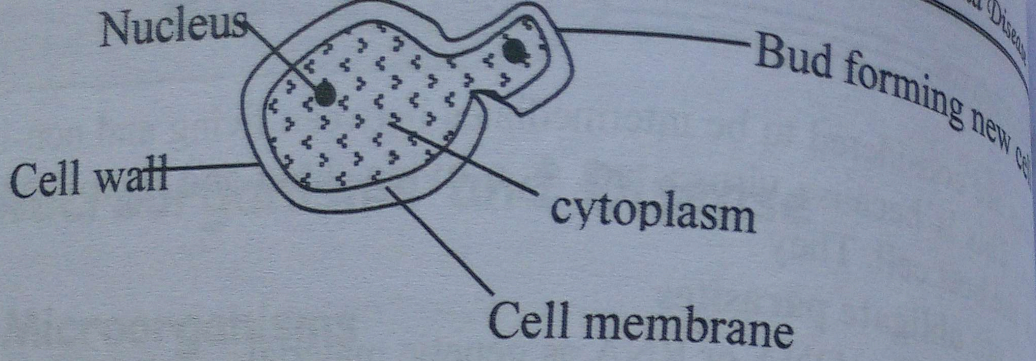


Fig. 4.1 Dividing yeast cell

ii. Moulds

Moulds are made up of filaments called **hyphae.** A mass of hyphae together forms a dense mass called **mycelium.**
They reproduce asexually by spore formation.

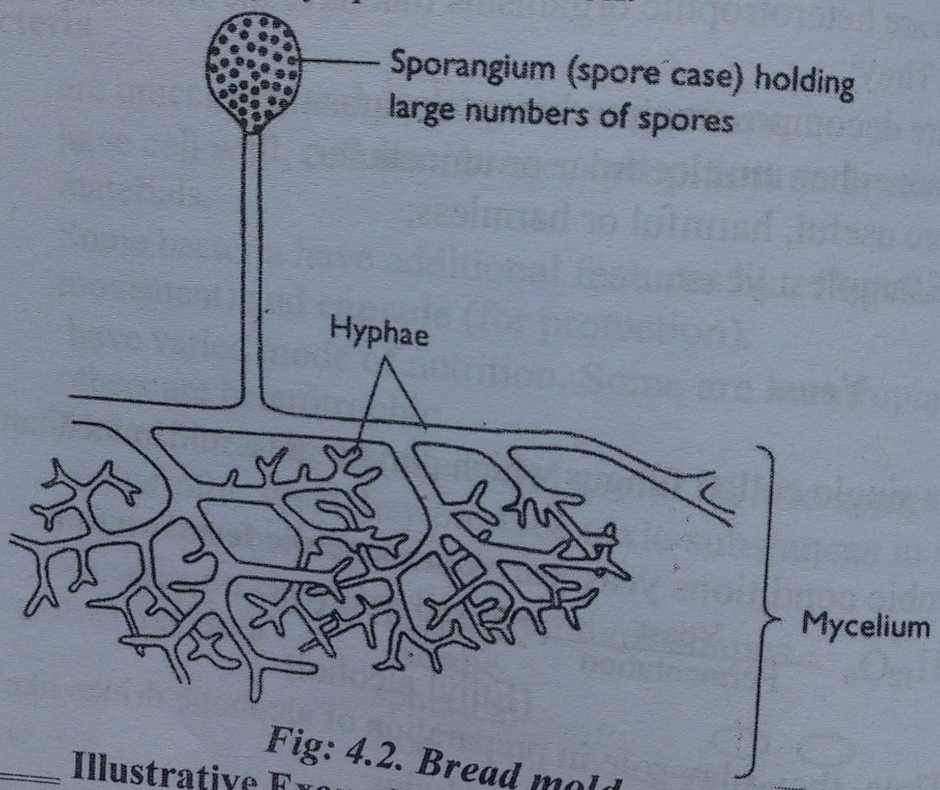


Fig: 4.2. Bread mold

Illustrative Examples

1. Which of the following is not commonly found in all bacteria
   A. Genetic material C. cell membrane
   B. Cytoplasm D. flagella
   Explanation: Bacteria's are prokaryotic single celled organisms.
   - They have cell wall, cell membrane, cytoplasm and genetic materials. Some bacteria's have additional extensions such as **flagella** and **capsules**.

   Answer: D

2. Which of the following is produced by bacterial action?
   A. Beer B. Wine C. Yoghurt D. Ethanol

Explanation: Beer wine and ethanol are produced by the fermentative effect of yeast. While cheese and yoghurt are prepared by the action of bacteria's on the lactose (milk sugar)

Answer: C

3. We express the size of viruses in (EGSECE 2001 E.C)
   A. Millimeter
   B. Micrometer
   C. Micro centimeter
   D. Nanometer

Explanation: Viruses are the smallest living organisms, ranging in size from about 20 – 3000nm; on average they are about 50 times smaller than bacteria.

Answer: D

4. Which of the following is correct about all bacteria? (EGESECE 2004 E.C)
   A. Presence of genetic material
   B. Presence of flagella
   C. presence of capsule
   D. similarity in shape

Explanation: All bacteria's have genetic materials.

- Flagella and capsule are only found in some groups of bacteria
- Bacteria vary in shape including; spherical, spiral, rod and comma shapes.

Answer: A

5. Which of the following types of bacteria are incorrectly matched with their shapes?
   A. Cocci – spherical
   B. Bacilli – Rod
   C. sprilli – spiral
   D. vibro – circular

Explanation: Vibrobacterias have comma shape

Answer: A

6. Bacteria are involved in all of the following except
   A. Lactic fermentation
   B. Making of vaccine
   C. alcoholic fermentation
   D. Tanning of lather

Explanation: Alcoholic formation is carried by unicellular fungi (yeast) under anaerobic conditions.

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow[\text{Fermentation}]{\text{Yeast}} \underset{\text{Ethanol}}{2C_2H_5OH} + CO_2 + ATP$$

Answer: C

7. Which of the following groups of microorganisms are not correctly matched with their reproductive mechanism?
   A. Bacteria – binary Fusion
   B. Virus – conjugation
   C. yeast – budding
   D. mould – sporullation

Explanation: Viruses do not exhibit sexual reproduction (conjugation)

- **Conjugation** is the simplest form of sexual reproduction

Answer: B

8. Describe at least two uses of bacteria:

Explanation: Bacteria are useful in the following major areas:

- Tanning of lather in lather industries
- Making of cheese and yoghurt
- Decomposition of dead bodies.
- They are used widely in the process of genetic engineering

9. Explain why viruses are called obligate parasites.

Explanation: Viruses are ultramicroscopic structures that only reproduce when they are inside the host cell. That is the reason why they are called obligate parasites.

Post Test 1

***Part I*: Choose the best answer from the given alternatives.**

1. Which one of the following is NOT a microorganism?
   A. Algae C. Fungi
   B. Bacteria D. Virus
2. A microorganism that is used in preparation of alcohol is:
   A. Mould C. Yeast
   B. Mushroom D. Penicillium
3. Which one of the following is an intermediate between living and non living thing?
   A. Bacteria C. Mould
   B. Virus D. Yeast
4. The bacterial action on milk to produce cheese and yoghurt is by:
   A. Citric acid C. Ethanoic acid
   B. Butyric acid D. Lactic acid
5. Which one of the following is untrue about fungi?
   A. They are parasitic
   B. They are decomposers
   C. They are authotrpohs
   D. They are free living
6. All of the following are structures common to all bacteria except:
   A. Cell wall C. Chloroplast
   B. Nuclear material D. Cytoplasm

7. Which o
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   B. V

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7. Which of the following types of bacteria is incorrectly matched with its shape?
   A. Cocci- spherical
   B. Vibro- comma
   C. Sprillum- spiral
   D. Bacilli- Zigzag

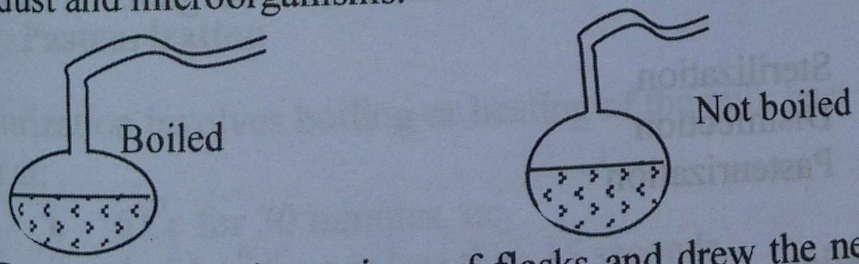
The germ theory of disease

Germs are groups of organisms that are responsible for the cause of some infectious diseases.
In the former times before the discovery of microorganisms people had various thoughts about the causes of disease. Some of these include.

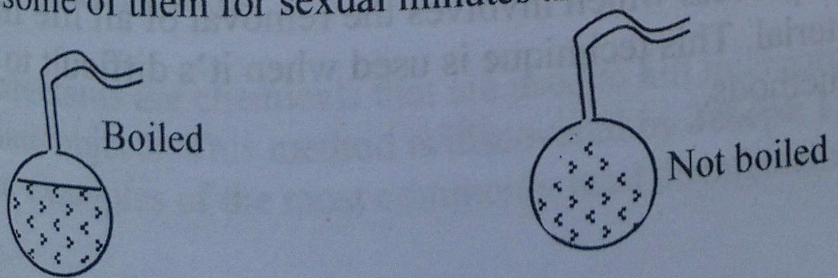
i. Anger by their God.
ii. Curse
iii. Evil sprit
iv. Bad air etc; however, the development of microscope by Antoine van Leeuwenhoek in 17th century helped different biologists to explain the relationship between infectious diseases and microorganisms.

The development of knowledge about microorganisms is actually related to the 'theory of spontaneous generation"

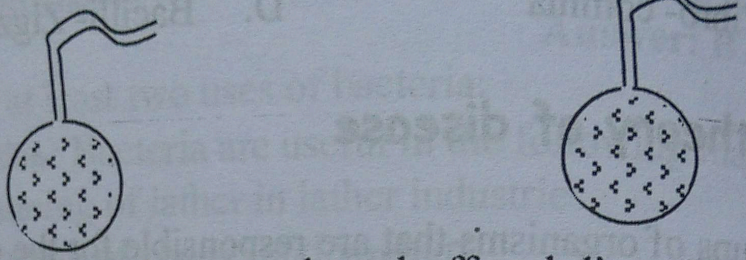
- ***The theory of spontaneous generation*** states that certain non-living things can be directly changed in to living things. This theory is opposed by many biologists and a French biologist Louis Pasteur disproved it finally by using an s-shaped flask that traps dust and microorganisms.



1. Pasteur prepared a serious of flasks and drew the necks out in to narrow curved extensions. Without sealing he boiled the broth in some of them for sexual minutes and left others unboiled.



2. All the flasks were left in calm air and within 36 hours unboiled flasks were full of microbes and moulds, while the boiled flasks remain unchanged.



3. If the curved necks were shaped off and dipped in to the broth of the boiled flasks, microbes immediately began to grow.
Louis Pasteur indicated that, infectious diseases are caused by microorganisms. He also introduced the technique of heating and boiling the milk and other food stuffs. This technique is called **pasteurization**

✓ **Pasteurization** – is a method of heating milk and other food stuffs which is followed by rapid cooling.

Robert Koch is the first person who demonstrated diseases are caused by certain bacteria. He showed for the first time that ***bacillus anthracites*** was the cause of a disease called **anthrax.**

Microbiological techniques

Microbiological is a method of controlling microorganisms since some of them are responsible for spread of disease.
The major techniques that are used to control microorganisms include:

i. Sterilization
ii. Disinfection
iii. Pasteurization

1) Sterilization

Sterilization is the process which involves the removal of all life from any object or material. This technique is used when it's difficult to obtain results by other methods.

Some methods that are used as sterilization include:

a) **Autoclave sterilization**

Autoclave sterilization is used to sterilize equipments which are not damaged by heat and for bacterial spores which do not die at lower temperature.

- It involves killing of bacterial spores by boiling them in water at 121^0c under high pressure for 14 – 45 minutes

b) **Ultra high temperature (UHT)**

UHT is a used to treat food to kill microorganisms found on it. This is done by heating the food within the temperature that ranges from 135^0c to 150^0c for 2 – 6 seconds.

- UHT I important because it is used in food processing industries where the food can be sterilized and do not get contaminated for years.

c) **Dry heat sterilization**

Dry heat sterilization is a method which used to sterilize scientific equipments.

- It requires high temperature and longer exposure time
- It involves:
  i. Heating for an hour at 171^0c or
  ii. Heating for 2 hours at 160^0c.

2) **Pasteurization**

Pasteurization involves boiling or heating of the milk beer and other food stuffs at:

i. 63^0c for 30 minutes, or
ii. At 71.6^0 for not less than 15 seconds.

- This process includes heating which is followed by rapid cooling.

3) **Disinfectants**

Disinfectants are chemicals that are used to kill microorganisms on in animate objects. This method is discovered by **Joseph Lister**
Some examples of the most commonly used disinfectants include:

- Dilute bleach and calcium hypochlorite – used to disinf... drinking water.
- Household bleach are used to keep sinks, toilet and ... containers and surfaces germ free.
- Phenol (carbolic acid) – used for disinfecting knives used f... surgery, operating table and clothes.
- Other examples of disinfectants include iodine, chlorine formalin, alcohols, etc.

Antiseptics are chemicals (disinfectants) that can be applied to a living tissues such as skin. They help to protect entrance of germs if the skin is cut or wounded.

Growing Microorganisms

Microorganisms can be grown in laboratories under controlled conditions. Growing microorganisms is important for various purposes. These includes

i. To know how to kill them
ii. To develop vaccines
iii. To identify their useful and harmful aspects
iv. To develop our biological knowledge, etc.

For growing microorganisms a biologist needs to fulfill the following preconditions.

1. Isolating the type of microorganisms to be studied.
2. Developing suitable nutrients and
3. Suitable environmental conditions.

Nutrient medium – a collection of nutrients required for growth of microorganisms.

- It can be solid or liquid
  - **Agar-** Is a solid nutrient medium which is extracted from red algae. It contains all the nutrients required to grow the microorganisms.
    - Microorganisms are cultured in a petridish
  - **Broth** – is a liquid nutrient medium.
    - Microorganisms are cultured in a test tube.

Culture media – is a collection of nutrients and microbial cells growing on a medium.

Inoculation – is a technique of transferring microorganism sample in to a nutrient medium.

✓ Inoculating needles should be sterilized **(Incinerated)** to avoid contamination

– A microorganism sample should be placed on the solid nutrient medium containing dish in a Zig-zag form to avoid competition between the microorganisms.

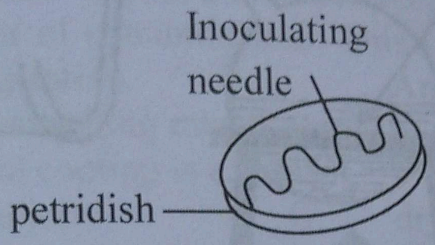


Figure 4.3: Innoculation

Simple stains for bacteria

Staining bacteria's helps us to see colored parts so as to make them easier to see and study.

- For example **crystal violet** stains all bacteria's purple, which makes it easier to identify their shapes.

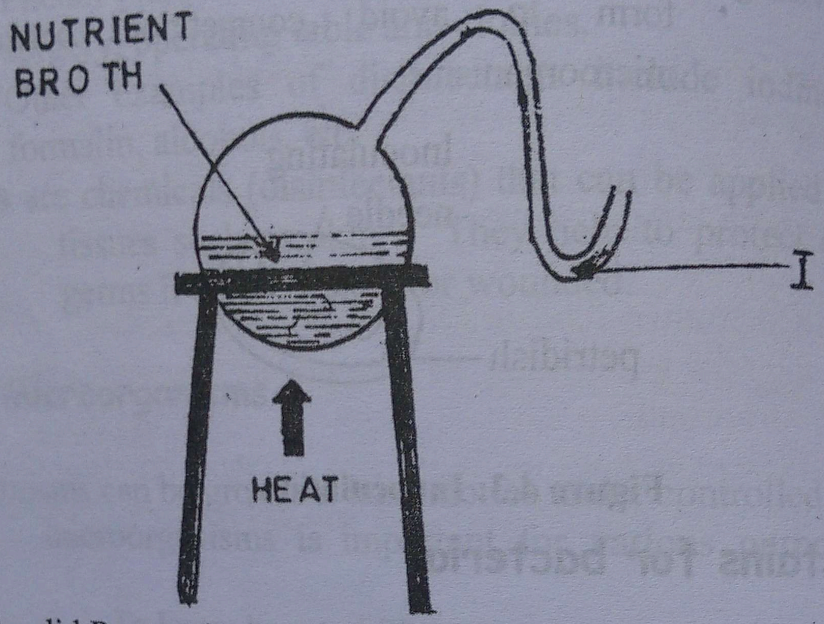
Gram staining technique helps us to identify between some groups of bacteria's. Gram staining follows the following procedures:

i. Stain all the bacteria with **crystal violet**
ii. Add **gram's iodine** and leave for about 1minute. it forms violet – iodine complex.
iii. Add **95% ethanol solution** to decolorize the stains.
iv. Some bacteria's get decolorized and other keep their original stain which is violet – iodine complex.
v. Add another stain **sufranin** and wash it with distilled water.

Accordingly those bacteria which retained (Kept) their primary stain and not decolorized are **Graham +ve bacteria's.** While those bacteria's that became decolorized are called Graham **–ve bacteria's.** All this happened because bacteria differ in the nature of their cell wall.

Illustrative Example

Question number 10 is based on the following diagram which shows Pasteur's famous flask with a neck bent into s-shaped curve.



10. Why did Pasteur bend the neck at curved – I?
    A. To trap dust and **microbes**.
    B. To make the flask air – tight.
    C. To enable the stream easily come out
    D. To make dust and microbes pass in and out freely

    Explanation:
    Louis Pasteur is a French biologist who disproved the theory of spontaneous generation. He disproved this theory by making an s-shaped flask and which traps the entering microbes and dust particles.

    Answer: A

11. If you want to sterilize a room by killing microorganisms and their spores, which of the following methods would you use?
    A. Chemical sterilization
    B. Irradiation
    C. autoclave sterilization
    D. Moist heat sterilization

    Explanation:Irradiation is a method of sterilizing rooms by using UV – rays in order to kill microbes. This method is mainly used in hospital rooms.

    Answer: B

12. If you want to introduce a small number of microorganisms in to a nutrient medium, which of the following would you use? (EGSECE 1996 E.C)
    A. Inoculation
    B. Irradiation
    C. sterilization
    D. pasteurization

Explanation: Inoculation is a method of introducing a sample of microorganisms into nutrient medium.

Answer: A

13. Heating, salting, drying and refrigeration are methods of (EGSECE 1998 E.C)
    A. Killing bacteria
    B. Preventing disease
    C. extracting antibiotics
    D. preserving food

Explanation: Preserving food involves keeping the food for a long time without fear of contamination. This includes heating, salting drying and refrigeration.

Answer: D

14. The technique controlling microorganisms which involves heating followed by rapid cooling is called
    A. Pasteurization
    B. Autoclave sterilization
    C. dry heat sterilization
    D. disinfection

Explanation: pasteurization is a method which is introduced by Louis Pasteur by heating and cooling milk wine etc.

Answer: A

15. Louis Pasteur's experiment on microorganisms using a flask with a lengthened s – shaped neck was to (GSECE 2001 E.C)
    A. Prove that germs cause diseases
    B. Kill pathogen using heat
    C. Prove the fermentation of sugar
    D. Disprove the theory of spontaneous generation.

Explanation:
The theory of spontaneous generation which states that certain non – living things can be changed in to a living thing was disproved by Louis Pasteur by using an s – shaped flask which traps dust and microorganisms from entering.

Answer: D

16. Which of the following is use of antiseptics?
    A. Sterilize rooms
    B. Clean kitchen utensils
    C. kill bacteria in the food
    D. kill bacteria on the skin

Explanation: Disinfection is a method of using chemicals to kill microbes on in animate objects. Antiseptics are disinfectants which are used for skin. Examples include iodine, alcohol etc.

Answer: D

17. Autoclaving kills microorganisms at
    A. 171^0c for 60 minutes
    B. 71.6 for 15 minutes
    C. 100^0c for 15 minutes
    D. 121^0c for 15 to 45 minutes

Explanation:

Microbiological technique	Temperature	Exposure
Autoclave sterilization	121^0c	15 – 45 minutes
Ultra high temperature	135^0c–150^0c	2 – 6 seconds
Pasteurization • Holding method • High temperature method	 63^0c 71.6^0c	 30 minutes 15 minutes
Dry heat sterilization	171^0c	1 hour
	160^0c	2 hours

18. Explain the meanings of:
    - Nutrient medium
    - Culture media
    - Inoculation

Explanation:

✓ **Nutrient medium** is the collection of nutrients that are used to grow micro organisms under controlled conditions.
Example: Agar, both.

✓ **Culture medium** is a nutrient medium on which micro organisms are growing.

✓ **Inoculation** is an introduction of a microorganism in to the nutrient medium.

19. List out some examples of disinfects that are commonly used.

Explanation: The commonly used disinfections are:
- Dilute bleach or calcium hypochlorite.
- Household bleach
- Phenol (carbolic acid)
- Iodine
- Chlorine
- Formalin, etc.

20. What is the contribution of the following biologists in developing the germ theory of disease?
    - Joseph Lister
    - Louis Pasteur
    - Robert Koch

Explanation:

- ***Joseph Lister***- introduced that disinfectants can remove the pathogenic microorganisms from in animate objects.
- ***Louis Pasteur*** – disproved the theory of spontaneous generation and stated that diseases are caused by microorganisms.
  He also introduced 'pasteurization' a microbial technique.
- ***Robert Koch*** – identified some pathogenic microorganisms with the diseases they cause.

Explain the main ideas of the theory of spontaneous generation.

Explanation: The theory of spontaneous generation states that, certain non-living things can be directly changed in to a living thing. This theory is supported by Aristotle and his followers. This theory is disproved by Louis Pasteur.

During inoculation of microorganisms on the petridish containing agar, the microorganisms are inoculated in a zigzag form. Explain why?

Explanation: Inoculation of microorganisms on a petridish in a zigzag form is to avoid competition between the microorganisms.

Why should we need to incinerate the inoculating needle during inoculation?

Explanation: Inoculating needles which are used during inoculation should be incinerated in order to avoid contamination.

Differentiate between Graham + ve and Graham –ve bacteria.

Explanation:

- Graham positive (+ve) bacteria that retain their primary stain crystal violet even by the decolorizing effect of ethanol.
- Graham negative (-ve) bacteria get decolorized their primary stain crystal violet up on the addition of ethanol solution which has decolorizing effect.

Post Test 2

I: Choose the best option from the given alternatives.

Who is the first to disclose microorganisms as causes of disease?

A. Louis Pasteur C. Robert Hooke
B. Robert Koch D. Joseph Lister

A solid nutrient medium that is made from algae is:

A. Agar B. Broth C. Iodine C. Dilute bleach

3. A transfer of microorganism in to a nutrient medium is:
   A. Staining
   B. Inoculation
   C. Pasteurization
   D. Disinfection
4. A theory which states that certain non living things can be directly changed in to a living thing is:
   A. Germ theory of disease
   B. Spontaneous generation
5. Chemicals that are used to kill microorganism on inanimate objects are:
   A. Antiseptics
   B. Antibiotics
   C. Disinfectants
   D. Antibodies
6. Which one of the following is an example of antiseptics?
   A. Dilute bleach
   B. Alcohol
   C. Phenol
   D. Formalin

Part II: Matching

Column A	*Column B*
1. Holding method of Pasteurization	A. 71.6^0c for 15 minutes
2. Dry heat sterilization	B. 121^0c for 15- 45 minutes
3. Autoclave sterilization	C. 135^0c–150^0c for 3-6 seconds
4. Ultrahigh sterilization	D. 63^0c for 30 minutes
5. High temperature method pasteurization	E. 160^0c for 2 hours

The immune system

Immune system is the system in which the body protects itself against disease causing microorganisms and foreign proteins. Our body has a natural ability to defend itself from diseases.

Immunity is the resistance of the body against disease.

Antigens are proteins which are new or foreign to the body. When antigens enter into the body, the body gets stimulated and gives a response against them.

- White blood cells (lymphocytes) produce special proteins called **antibodies** which fights against the foreign proteins (antigens) carried by pathogens. In addition **phagocytes** are another types of white blood cells engulf and digest the weakened proteins.

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  - Skin wh
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  - Cilia in
  - Mucus
    linings
  - HCl in
  - White

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Immunity

Natural (Inherited) immunity

- This is the natural ability of the body to fight disease producing organisms.
- It involves protective abilities of
  - Skin which is barrier to germs
  - Cilia in the nose
  - Mucus in the epithelial linings
  - HCl in the stomach
  - White blood cells.

Artificial (acquired) immunity

- This type of immunity is used when the pathogens go beyond the resistance of our immune system
- It involves
  - Vaccinations
  - Antibiotics

Antibiotics

Antibiotics are chemicals which are used to kill bacteria inside our body. These chemicals can kill bacteria's without damaging our body.

The first model antibiotic is penicillin which is introduced by Sir Alexander Fleming in 1928. It is made from a fungal species called ***Penicillium notation***.

The other types of chemicals that are used to cure fungal diseases are called **antifungal chemicals.** These are similar to antibiotics but kill fungal cells.

However, viral diseases have not yet been treated by using any kind of drug. Thus medical scientists are working so hard to develop effective drugs against viral diseases.

Artificial Immunity

Our body has its own natural ability to protect itself against disease. However, if the immune system of our body fails to defend some dangerous diseases, it will be treated or protected by artificial (acquired) immunity.

- ✓ Artificial immunity is given in the form of **vaccine or drugs**.
- ✓ Artificial immunity can be **active** or **passive** immunity.

i. ***Active Immunity*** – involves introduction of weakened or dead pathogens in the body which it stimulates the body to produce its own antibodies.
   - ✓ It has long lasting effect on the body.
   - ✓ Children should be –vaccinated for some diseases early age.

iii. ***Passive immunity*** – it involves giving specific antibodies the form of injection
   - ✓ It can also be natural form mother to child until the child produces its own **natural active immunity.**
   - ✓ It provides a high type of resistance but lasts only for short time

Vaccination (Immunization)

Vaccination is the use of dead or weakened forms of pathogens to produce immunity (antibodies) to dangerous diseases.

The vaccination works through the following pathways.

1. Weak or dead form of the infectious microorganisms (pathogens) is put in to the body inform of injection or through the mouth.
2. Once the wakened or dead pathogens are in the body, the white blood cells respond by producing antibodies.
3. If the living microorganism enters the body in the future, antibodies are produced very rapidly to destroy it and so the diseases do not develop.

Thus, our country Ethiopia is strongly working to increase the number of children who are immunized against the most common diseases like tetanus, polio, tuberculosis, whooping cough, diphtheria, measles, etc.

The beginning immunization is accredited to an English doctor who named **Edward Jenner** who developed vaccination against small pox (ፈንጣጣ).

Note: Small pox is a viral disease that is currently eradicated from the world.

Immunity

Natural

Natural active	Natural passive
➢ Ability of the body to produce its own antibodies to fight disease without using any kind of external stimulating factor	➢ From mother to child, until the baby develops its own antibodies

Artificial

Artificial active	Artificial passive
➢ Introducing dead or weekend form of pathogens in to the body in order to stimulate the body to produce its own antibodies.	➢ Introducing antibodies in to the body in which the body has no role in producing antibodies.

Illustrative Example

25. How does immunization against diseases such as diphtheria and polio limit the spread of infectious diseases? (EGSECE 1996 E.C.)
    A. Immunization kills the relevant pathogens.
    B. Immunization suppresses or reduces the immune response and associated inflammation.
    C. Immunization strengthens first – line defense barriers and prevents the entry of the relevant pathogens in to the body.
    D. Immunization reduces the multiplication of the pathogens in immunized hosts.

Explanation:
Immunization programs for diseases such as diphtheria and polio is an active immunity in which the body produces its own antibodies to inhibit the attack of pathogens

Answer: C

26. The treatment of diseases by chemicals that kill or inhibit the growth of microorganisms without harming the host at that concentration is:
    A. Immunization
    B. Chemotherapy
    C. physiotherapy
    D. sterilization

Explanation:
Immunization is the process of introducing dead or weake forms of pathogens into the body at the concentration in whic cannot harm the body.

Answer:

27. A chemical that destroys or weakens disease – causing microorganisms is known as. (EGSECE 1997)
    A. An antibiotic
    B. A toxin
    C. An endospore
    D. A vaccine

Explanation: antibiotics are chemicals that are used to kill weaken pathogenic microorganisms. **Answer: A**

28. Many vaccinations stimulate the immune system by exposing it to
    A. Weakened microbes
    B. Antibodies
    C. enzymes
    D. mutated genes

Explanation: Artificial active immunity is a method of introducing weakened or dead microbes in to the body in order to stimulate the immune system of the body to develop its own antibodies.

Answer: A

29. During the last 50 years, over- use of prescription drugs has led to the emergence of resistant strains of pathogens. Why is this a problem? (EGSECE 1998 E.C)
    A. Many diseases may become untreatable
    B. Resistant pathogens will cause new diseases.
    C. A single prescription drug can no longer kill all strains of a pathogen.
    D. Prescription drugs will cause the release of toxins by pathogens.

Explanation:
Using the same drug over again and again causes the pathogens to develop adaption of resisting the drugs.

30. Children are immunized against the six target diseases. A child who is immunized could not be infected by
    A. Hook warm
    B. Amoeba
    C. typhoid
    D. polio

Answer: C

Explanation:
Immunization against six targeted in Ethiopia was child vaccines against the following diseases. These are Diphtheria, pertosis, tetanus, measles, Tuberculosis, polio.

Answer: D

31. Which of the following is true of vaccines? (EGSECE 2004 E.C)
A. Vaccines are alive and activated
B. Vaccines cause diseases.
C. Vaccines protect the body from disease
D. Vaccines weaken the immune system

Explanation: Vaccines stimulate the body to develop its immune capacity against pathogenic microorganisms. **Answer: C**

32. The kind of immunity that is gained as a result of vaccination with weakened or dead pathogens is:
A. Passive immunity
B. Natural immunity
C. Inborn immunity
D. Active immunity

36. All of the following are chemical approaches to control microorganisms EXCEPT. (EGSECE, 2005 E.C)
A. Antibiotics
B. Disinfectants
C. antiseptics
D. autoclaving

Explanation: Microbiological technique is a technique of controlling the spread of microorganisms.
- The major microbiological techniques are.
  - Sterilization – applying heat to kill or eliminate microorganisms.
  - Disinfectants – use of chemicals to remove microorganisms on in animate objects
  - Pasteurization – a method of boiling food items like milk which is followed by rapid cooling.
  - Antisepts – chemicals used for living bodies like skin.

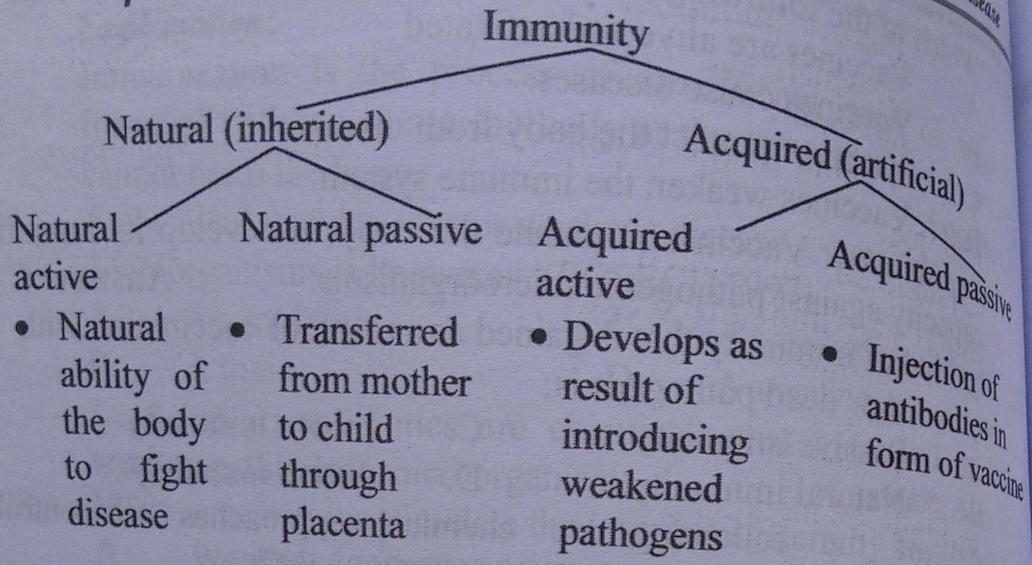
Answer: D

37. How antiseptics do used to destroy microorganisms? (EGSECE, 2005 E.C)
A. By dry heat application in special yens.
B. By applying them on cut skin
C. By apply them on in animate objects.
D. By taking them in to the body.

Explanation: Antiseptics are chemicals that are used on a living tissues like skin. **Answer: B**

38. Natural active immunity is an immunity which we: (EGSECE, 2005 E.C)
A. Get from vaccines after being infected by the pathogen.
B. Develop as a result of vaccination.
C. Develop by once being exposed to the pathogen.
D. Get from our mother through the placenta or breast feeding.

Explanation:



Answer: C

Explanation: Active immunity is a kind of immunity which involves introduction of dead or weakened forms of pathogens in to the body in order to make it get stimulated to fight disease.

33. Distinguish between:
    - Inherited and acquired immunity
    - Active and passive immunity

Answer: D

Explanation: The difference between inherited and acquired immunity is:

Inherited (natural) immunity	Acquired (artificial) immunity
• It is the natural ability of the body to fight disease. • First line of defense includes skin, mucus, cilia in the nose, HCl in the stomach, etc. • Second line of defense involves phagocytes and lymphocytes which produce antibodies.	• They are immunities that are taken in the form of injection or through the mouth. • It can be active when weakened or dead pathogens are introduced in to the body. • It can be passive when antibodies are injected in to the body.

✓ **Active immunity** – involves introduction of weakened or dead pathogens in to the body which stimulates the body to produce its own antibodies. It has long lasting effect.

✓ **Passive imm**
in which th
antibodies.

34. List out some of t
Ethiopia.
Explanation: In E
children are immun
➢ Tetanus
➢ Polio
➢ Tuberculosi

35. Explain the meani
• Immunity
• Pathogen
• Antibody
• Antigen

Explanation:
✓ **Immunity**
✓ **Pathogens**
✓ **Antibodie**
body to fig
✓ **Antigens**

Part II: *Choose the bes*

1. A fungal specie
antibiotic is:
A. Penicilliu
B. Mushroo

2. Among the foll
one is NOT the
A. Antibody
B. HCl in th

3. Proteins produc
A. Antigens
B. Antibod

4. Which one of t
A. Involves
B. It can al

✓ **Passive immunity** – introducing antibodies in to the body in which the body has no role in the production of antibodies.

34. List out some of the diseases which children are immunized in Ethiopia.

Explanation: In Ethiopia, the most commonly diseases which children are immunized (vaccinated) for are:

- Tetanus
- Polio
- Tuberculosis
- whooping cough
- Diphtheria
- Measles

35. Explain the meanings of:

- Immunity
- Pathogen
- Antibody
- Antigen

Explanation:

✓ **Immunity** is the resistance of the body against disease.
✓ **Pathogens** are disease causing microorganisms.
✓ **Antibodies** are special proteins that are produced by the body to fight pathogenic microorganisms.
✓ **Antigens** are proteins that are foreign or new to the body.

Post test 3

Part II: Choose the best answer from the given alternatives.

1. A fungal species that was first used to make the first model antibiotic is:
   A. Penicillium notatum
   B. Mushroom
   C. Candida albicans
   D. Clostridium tetani
2. Among the following defensive mechanisms of the body, which one is NOT the first line of defense?
   A. Antibody
   B. HCl in the stomach
   C. cilia in the nose
   D. mucus
3. Proteins produced by the body to fight against foreign proteins is:
   A. Antigens
   B. Antibody
   C. Antibiotics
   D. Antiseptics
4. Which one of the following is untrue bout passive immunity?
   A. Involves giving specific antibodies in the form of injection
   B. It can also be natural form mother to child

C. Involves introduction of weakened or dead pathogens into the body
D. lasts only for a short time

5. Which one of the following is acquired immunity?
A. Antibiotics C. Cilia
B. Skin D. Mucus

Part II: Matching

Column A	*Column B*
1. Immunity	A. Disease causing microorganisms
2. Antigen	B. are chemicals which are used to kill bacteria inside our body
3. Pathogens	C. The ability of the body to fight against disease
4. Antibiotics	D. Proteins that are foreign to the body
5. Antibodies	E. Special proteins that are produced by the body to fight against disease

4.2. Diseases

Disease is any form of abnormality or disorder in or on the body that distorts its normal functioning
Diseases can be caused by infections microorganisms or by some other causes like deficiency diseases, genetic disorders etc.
Some of the most commonly known diseases which are caused by pathogenic microorganisms will be discussed as follows.
Pathogens – are parasitic microorganisms that cause diseases.

1) Bacterial diseases

Most groups of bacteria are harmless; while, some groups of bacteria are harmful.
Harmful bacteria are pathogenic. This includes:

a. Tuberculosis (TB)

TB is caused by a bacterium called ***Mycobacterium tuberculosis***. It can be easily transmitted by droplet infection.
- ✓ It can spread easily to those people who have weak immune system.

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b. **Cholera**

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- **Symptoms** – loss of weight and appetite
  - persistent cough
  - loss of weight.
  - sputum with the blood (occurs sometimes)
  - slight fever
- **Control and prevention** – avoiding crowded living environment
  - good nutritional state.
  - good ventilation.
  - personal hygiene and environmental sanitation.
  - pasteurization of milk.
  - effective use of drug and vaccination.
  - Covering the nose and mouth during coughing and sneezing.

TB is a very common problem in Ethiopia. Its spread is related to

i. The spread of HIV/AIDS
ii. Development of resistance varieties of bacterium against drugs.

b. Cholera

Cholera is caused by a bacterium called ***Vibro cholera.*** It is a bacterium infects the intestine.

- **Transmission** – eating or drinking contaminated food and water with fecal waste.
- **Symptoms** – sometimes asymptomatic (no recognizable symptom).
  - Diarrhea, vomiting and dehydration.
  - Pale, watery diarrhea is severe symptom that can kill.
  - Muscle scramps
- **Control and prevention** – taking more fluid in order to replace the fluid lost in the form of diarrhea.
  - Using antibiotics and vaccines.
  - Proper disposal of feces.
  - Treatment of drinking water.
  - Good personal hygiene and environmental sanitation.
  - Avoiding consumption of uncooked food.

Cholera infection can affect largely travelers, people who are in some sort of disaster, festival or big celebration. It is also very common problem in Ethiopia

c. Typhoid

Typhoid is caused by a bacterium called ***Salmonella typhi***. It affects intestinal tract and sometimes the blood stream.,

- **Transmission** – polluted water, food, vegetables which are infected with feaces and urine of infected person.
- **Symptoms** – abdominal pain, sore throat, headache, high fever, diarrhea, spots on chest and abdomen, enlargement of spleen.
- **Control and prevention**
  - Proper disposal of feaces
  - Treatment of water
  - Cocking foods and vegetables.

The status in Ethiopia is very common.

2) Protozoan diseases

a) Malaria

Malaria is caused by a protozoan called **Plasmodium.** It is transmitted from infected person to a healthy person by the bite of parasitized female anopheles mosquito. Thus, anopheles mosquito is a **vector** for malaria.

Vector – is an agent for transmission of disease.

Plasmodium parasitizes the blood and liver of an infected person.

- **Symptoms** – fever, chills and sweats.
  – It makes people unable to work.
- **Control and prevention** – Avoid contact with mosquitoes by using net over beds, fixing screens on windows, using mosquito coils and repellants.
- Minimizing opportunities of mosquitoes to breed by draining swamps or covering water tanks
- Killing adult mosquitoes by insecticides
- Biological control by using fish that feed on mosquito larvae.

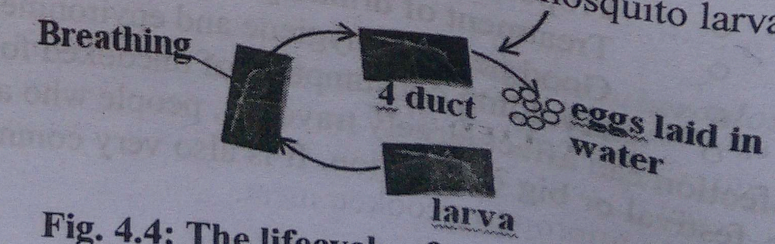

Fig. 4.4: The lifecycle of plasmodium

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Malaria kills millions of people each year in sub-Saharan African countries. Therefore, malaria is a very common disease in Ethiopia.

b) Amoebic dysentery

Amoebic dysentery is caused by a protozoan species called ***Entamoeba hystolytica***

- It is transmitted by eating and drinking contaminated food and water. And also house flies may infect the food and water.

Symptoms

- Ulceration of large intestine.
- Diarrhea with loss of blood, abdominal pain.

Prevention and control

- Proper disposal of feaces, effective treatment of drinking water, prevent infective food handles like house flies. Good personal hygiene and sanitation, hygienic food handling

Amoebic dysentery is a very common disease in Ethiopia.

3. Parasitic worms

Parasitic worms are known to cause various diseases to humans. They can be **round worms** such as **Ascaris**, or **flat worms** like tape worms. Among the most commonly known parasitic diseases, tape warm, ascaris, blood flukes, hook worm are common in Ethiopia.

Tape worm

Tapeworm is a flat worm that parasitizes the wall of intestine of humans. The most common type of tape worm in Ethiopia is ***Taenea saginata*** which uses cattle as intermediate host.

- ✓ **Transmission** – Eating improperly cooked meat or raw meat
- ✓ **Symptoms** – segments of tape worms in the feaces, weight loss, seizures, feeling of weakness due to pernicious anemia

N.B. pernicious anemia – is caused due to the lack of vitamin B_{12}.

- ✓ **Control and Prevention** – using anti-worm drugs, proper disposal feaces, efficient inspection of meat, avoid eating raw meat and improperly cooked meat.

- **Life cycle**
  i. ***Onchospheres*** – are structures that are released from segment and penetrate through the walls of small intestine the cattle and then to their blood, fatty tissues and muscl They develop in to **cysticerici.**
  ii. The cysts develop in to **scolex** which has suckers and hooks attach itself to the wall of the intestine.
  iii. The raw meat which contains the cysts attaches in to the wall intestine of human and begins to grow. After it attaches itself to the intestinal wall it starts develop new segments call **proglottids.**
  iv. ***Proglottids*** have male and female organs.

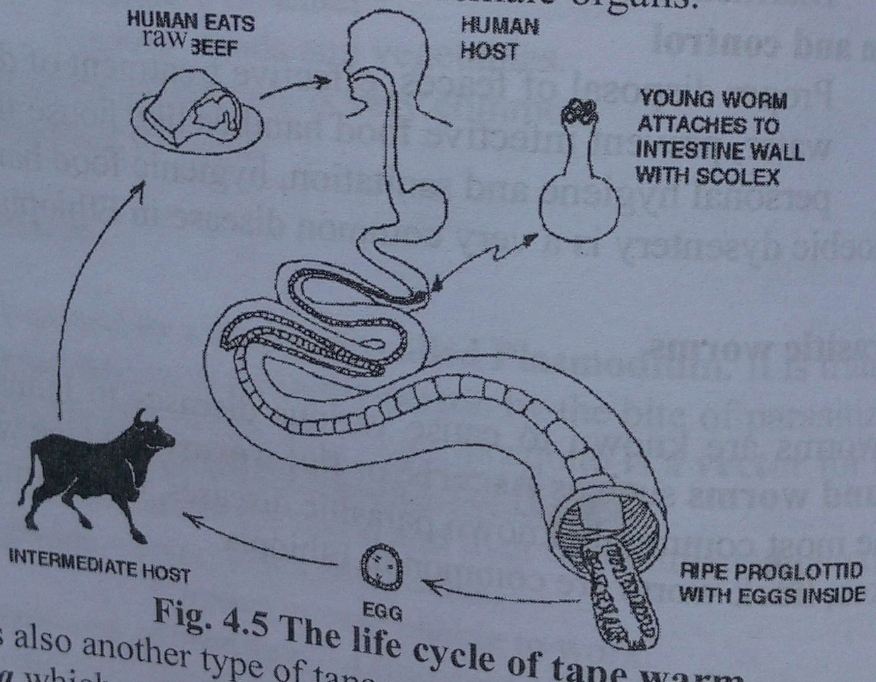

Fig. 4.5 The life cycle of tape warm

There is also another type of tape worm which is known as **Tanea *saginata*** which uses pig as intermediate host.
Tape worm is very common in Ethiopia. This might be related to our traditional practice of eating raw meat and problems related with inspection of meat.

4. Gastroenteritis

Gastroenteritis is an intestinal infection which is caused by different groups of pathogenic microorganisms such as viruses (rotaviruses), bacteria (Salmonella and *Escherichia coli*), Amoeba etc.

✓ It's characterized by **acute watery diarrhea** (AWD). It may result in excessive loss of the body fluid.

- ***Symptoms*** – w
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- ***Symptoms*** – watery diarrhea, slight fever, abdominal pain, muscle ache and headache.
- ***Control and prevention*** – good personal hygiene (washing hands thoroughly), treatment of water, cleaning food and equipments, avoid eating under cooked or raw food, good nutritional state for the patients of AWD.

Gastroenteritis is a very common problem in Ethiopia

Illustrative Example

39. Which of the following is correct about tuberculosis?
   A. It affects only children
   B. It is transmitted by mosquito
   C. It is caused by virus
   D. It causes persistent cough

Explanation: Tuberculosis is a bacterial disease which is caused by *mycobacterium tuberculosis.*

- It can be easily transmitted by droplet infection, unpasteurized milk.
- Its symptoms include persistent cough, slight fever, loss of weight etc.

Answer: D

40. Diarrhea is the symptom of all of the following diseases except: (EGSECE 2004 E.C)
   A. Gastroenteritis
   B. Typhoid fever
   C. cholera
   D. malaria

Explanation: The symptoms of malaria include high fever, headache, heavy sweating, shivering and chills

Answer: D

41. What is the role of vectors in relation to the health of organisms? (EGSECE 2004 E.C).
   A. They carry disease causing organisms from one host to another
   B. They are causative agents of many diseases.
   C. They prevent micro-organisms from infecting healthy individuals.
   D. They deactivate diseases causing micro-organisms.

Explanation:
Vectors are agents that transmit or carry pathogenic organisms from infected person or contaminated objects to healthy person.

Answer: A

42. The famous plant "endod" (soapberry plant) helps to fight a serious human parasitic disease known as:
A. Filariasis C. tape worm
B. Amoebiasis D. bilharziasis
Explanation: "Endod" is an indigenous plant which is used to eradicate snail which is an intermediate host for bilharzias.
Answer: D

43. Which of the following human diseases is caused by a virus?
A. Cholera B. Polio C. Anthrax D. Malaria
***Explanation*:** Cholera and Anthrax are caused by bacterium.
- Malaria is a protozoan disease.
- Polio is a viral disease which is caused by polio virus

Answer: B

44. Which of the following disease controlling methods works for malaria?
A. Proper disposal of feaces
B. Hygienic food handing
C. prevention of flies
D. killing adult mosquitoes
Explanation: Malaria is transmitted from an infected person to a healthy person by the bite of female anopheles mosquitoes, therefore eradication of anopheles mosquito is the best way to control the spread of malaria.
Answer: D

45. Which of the following human parasites need an intermediate host to complete its life cycle?
A. Tape worm C. leech
B. Ascaris D. hook worm
Explanation*:** There are various types of tape worms, among them the most common ones are ***Taenea saginata and Tenea solum. Intermediate host for ***Tanea saginata*** is cattle, while an intermediate host for ***Tanea solum*** is pig.
Answer: A

46. Which of the following disease is NOT carried by vectors?
A. Malaria C. ascaris
B. Filariasis D. yellow fever
Explanation: Ascaris is transmitted by contaminated food, water, and poor hygiene.
Answer: C

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47. Which method of controlling populations of mosquitoes most likely involves the least risk of causing effect to the environment?
A. Spraying adult mosquitoes with pesticides from air planes
B. Draining swamps where mosquitoes deposit eggs
C. Spraying oil on wet areas where mosquitoes breed
D. Releasing more predators of mosquitoes native to mosquito habitats.

Explanation:
Among various methods of controlling insects, the biological method which involves releasing more predators of mosquitoes is the best way since it has no or less effect on the environment.

Answer: D

48. Which of the following is the best method of controlling tetanus? (EGSECE 1997 E.C)
A. Vaccination
B. Avoid droplet infection
C. Avoid contact
D. Isolation

Explanation:
Clostridium tetani are bacterias that causes tetanus. They enter into the body through the wounded or cut skin. Therefore its difficult to control easily. The only way is using vaccines. **Answer: A**

49. In order to prevent himself from tape worm disease, what advice would you give to your friends?
A. To spray his house with insecticides
B. To be vaccinated
C. To avoid eating raw meat
D. To avoid crowded places.

Explanation:
Tape worm can be easily transmitted by eating raw meat or properly uncooked meat. **Answer: C**

50. Which of the following diseases is caused by fungus?
A. Amoebic dysentery
B. Ring worm
C. Trypanosomiasis
D. Malaria

Explanation: Amoebic dysentery, Malaria and Trypanosomiasis are protozoan diseases. Ring worm is a fungial disease.

Answer: B

51. Human parasitic worm which uses snail as its intermediate host is: (EGSECE 1999 E.C)
A. Hook worm
B. Ascaris
C. Tape worm
D. Bilharzia

Explanation: Intermediate host is a host in which the parasite begins it is life cycle up to some stage of development. For example bilharzias worm starts it is life cycle in snails which is completed in humans.

52. Crowded places should be avoided during an influenza epidemic to prevent
    A. Contact with mosquitoes
    B. Droplet infection
    C. contact with contaminated clothes
    D. picking up infected fleas

    Explanation: Influenza can be easily transmitted by droplet infections.

    Answer: B

53. Which of the following diseases is transmitted through direct contact over a long period?
    A. Diphtheria
    B. Typhoid
    C. Tetanus
    D. Leprosy

    Explanation: Leprosy is a bacterial disease which is caused by a bacterium called ***Mycobacterium leproae***. It is transmitted through contact over a long period of time.

    Answer: D

54. Which of the following methods is used to prevent the spread of amoebic dysentery?
    A. Eradication of water snails
    B. Eradication of anopheles mosquito
    C. Proper disposal of feaces
    D. Using net over bed

    Explanation*:** Amoebic dysentery is a protozoan disease which is caused by Entameoba hystolytica.***
    - Its spread is controlled by proper disposal of feaces, cooking the vegetables, washing hands before and after meal, etc.

    Answer: C

55. How an individual does knows that he is infected by tape worm?
    A. Observing tapeworm segments in feaces
    B. Observing tapeworm segment in urine
    C. Making blood test
    D. Making sputum test.

    ***Explanation*:**
    The symptom of tapeworm is appearance of segments in the feaces.

    Answer: A

Part I: Math the fol
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1. Tuberculosis
2. Tapeworm
3. Malaria
4. Amoebic dy
5. Typhoid
6. Cholera

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Part I: Math the following human disease with their causes.

Column A	*Column B*
1. Tuberculosis	A. *Tanea saginata*
2. Tapeworm	B. *Salmonela typhi*
3. Malaria	C. *Vibro cholera*
4. Amoebic dysentery	D. *Mycobacterium*
5. Typhoid	E. *Plamodium*
6. Cholera	F. *Entamoeba hystolytica*

Sexually Transmitted Diseases (STD'S)

STDS are diseases that can be transmitted from an infected person to a healthy person by unprotected or unsafe sexual contact. They are also known as **venereal diseases (VD).**

Most sexually transmitted disease are very dangerous because they are asymptomatic (have no any recognizable symptom).
STDs may cause severe damage to different organs of the body including heart and brain damage, even infertility.

Even though STDs can affect people at any age, the most vulnerable age group are younger people who are in between the ages of 15 – 30 since people at these age are sexually active.
The commonly known STDS are gonorrhea, chancroids, syphilis, HIV/AIDS.

1) Gonorrhea

Gonorrhea is caused by a bacterium called ***Neisseria gonorrhea.*** The bacteria are found on the mucus areas of the body such as vagina, penis, throat and rectum. It is common in people who have many sexual parents.

- **Transmission** – It can be transmitted by vaginal, anal or oral sex with an infected person. Other routes of transmission are:
  - Having un protected sex.
  - From mother to child during birth.
- **Symptoms**
  - In men the symptoms of gonorrhea are:

- burning sensation while urinating,
- yellowish white discharge from the penis. The symptoms appear within a week after infection.

- In women it is mostly asymptomatic, but sometimes it show symptoms like:
  - discharge from the vagina.
  - burning sensation while urination.

If the pregnant woman remains untreated gonorrhea, it can be transmitted to her baby and may result in the blindness of her baby.

➢ **Control and Prevention**

At early stages gonorrhea can be treated by using antibiotics. However it can be most effectively prevented by:

✓ Respecting the ABC rules.
A = Abstain yourself from sexual intercourse
B = Be faithful to one sexual partner.
C = use condom.

✓ Visit the clinics or hospitals if you suspect yourself that you are infected.

✓ If you are infected, do not have sexual intercourse until your treatment is completed.

2. Syphilis

Syphilis is one of the most serious STD. It is caused by a bacterium called ***Treponema palladium***. Just like any other STDs it is very common among those people who are at young age.

Transmission – un protected sexual intercourse
- Congenital syphilis spreads from the mother to the fetus and causes severe complications to the new born baby.

Symptoms – syphilis has three major stages of symptom. These are:

i) **Primary stage** (the first six weeks)
➢ Painless sore at the site of initial infection. This might be penis, entrance of vagina, in the mouth rectum or in the vagina.
➢ Swollen glands.

ii) **Secondary stage** (occurs six weeks after the primary stage)
➢ Tiredness, fever, sore throat, headaches, hoarseness, loss of appetite, patchy hair loss, swallows glands, rash.
➢ Lasts for two to six weeks.

iii) **Tertiary stage (La**
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iii) **Tertiary stage (Late syphilis)**

i. It appears after years and may cause severe, irreversible problems that involve illness of the skin, bones, central nervous system (brain and spinal cord) and heart.

ii. It can be transmitted from an infected pregnant woman to her baby and may cause death or deformity of the child.

Control and prevention

- Respecting the ABC rules
- Visit a clinic, health center or hospital in your locality if you think you are infected.
- Prenatal blood test for syphilis for pregnant women.
- If you are infected, avoid having sexual contact until the course of your treatment is completed.

3. Chancroids

Chancroid is a bacterial disease which is caused by a bacterium called *Hemophilus ducreyi*. It is the most common problem of men. It also increases the risk of HIV/AIDS infection.

Symptoms

- **First stage – involves:**
  - Ulceration on the genitals, particularly on penis which are filled with soft pus.
  - Bleeding of ulcers and bursting of pus.
  - Painful ulcer.
- **Second stage involves:**
  - Swollen glands which are filled with pus. When they brust it may cause large and painful ulcer.
  - Permanent loss of penis or groin.

Control and Prevention

It can be prevented early by taking effective dose of antibiotics. Moreover, it can be prevented through the following important steps:

- Respecting the ABC rules.
- Good genital hygiene
- Circumcision of males reduces the risk of chancroid.

Visit clinics or health centers in your locality if you think that you are infected.

Using medicines

- **Traditional practices** – are very important and alternative forms of medicine, especially in developing countries like Ethiopia. They do have holistic effect on treating the patient; however, limiting their dose is – very important.
- **Modern medicines** – are responsible for curing various diseases that have been a problem for a long time.
  - ✓ The most common examples of modern medicines include antibiotics and use of vaccines. However, care should be taken when we use modern medicines, so the following precautions should be considered;
    - Do not take more than you are prescribed
    - Do not take less than you are prescribed
    - Make sure you finish taking all the medicine
    - Do not use drug which is not prescribed for you.
    - Carefully observe the expiry date of the drug before use.

Illustrative Example

Item number 37 is based on the following table which shows a disease, and its prevention method

Disease	Prevention
Gonorrhea	Be faithful to partner
Syphilis	Use condom
Cholera	Y
Typed	Good personal hygiene

56. Which of the following represents 'y' in the table? (EGSECE 2004 E.C)
A. Avoid blood contact
B. Use condom
C. clean drinking water supply
D. avoid mosquito

Explanation:
Cholera can be easily transmitted by eating or drinking contaminated food or water. Thus it can be prevented by drinking clean water and eating properly cleaned and cooked food.

Answer: C

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57. Suppose your friend tells you that he suspects he is HIV positive. How would you react to this? (EGSECE 2004 E.C)
A. Avoid your friendship to protect yourself
B. Tell him to buy and take anti retroviral drugs.
C. Inform other friends that he is HIV positive
D. Advice him, to go to health center to know his status.

Answer: D

58. Human disease caused by bacteria include
A. Influenza C. rabies
B. Tetanus D. malaria

Explanation: Influenza and rabies are viral diseases.
- Malaria is caused by a protozoan species.
- Tetanus is a bacterial disease which is caused by *clostridium tetani*

Answer: B

59. Which one is primarily attacked by the human immunodeficiency virus (HIV)?
A. The muscle cells C. the white blood cells
B. The platelet D. the red blood cells

Explanation: HIV attacks primarily the white blood cells in order to weaken the immune system of the body. **Answer: C**

60. Which of the following sexually transmitted diseases can damage every organ in the body including the brain if not treated properly?
A. Syphilis C. chanceries
B. Trichomoniasis D. Gonorrhea

Explanation: Syphilis has three major stages. These are primary syphilis, secondary syphilis and tertiary (late) syphilis.
- The tertiary syphilis is the final stage where bacterium spreads through the body including the brain.

Answer: A

61. Which of the following sexually transmitted disease is caused by protozoa?
A. Syphilis C. Trichomoniasis
B. Chancroid D. Gonorrhea

Explanation:

STD	Cause
Syphilis	*Treponema pallidum*
Chancroid	*Hemophilus ducrey*
Trichomonasis	*Trichomonas*
Gonorrhea	*Neisseria gonorrhoeae*

Answer: C

62. Which of the following is true about cholera? (EGSECE, 2005 E.C)
A. It causes severe dehydration
B. Its common symptom is fever.
C. It is caused by viral infection
D. It spreads by contaminated air inhalation.

Explanation:
- *Cholera* – is a intestinal problem which is caused by a bacterium called *Vibro cholera.*
  - Its symptoms includes, diarrhea, dehydration.
  - It is transmitted by eating and drinking contaminated food and water.

Answer: A

63. Malaria causing single celled parasitic protozoan is called: (EGSECE, 2005 E.C)
A. Paramecium
B. Salmonella
C. Mosquito
D. Plasmodium

Explanation:
- **Malaria** is one of the diseases that kill many people in Africa. This disease is caused by a protozoan species called **plasmodium** which is transmitted from an infected person to a healthy person by the bite of Female anopheles mosquito.

Answer: D

64. A grade 9 student wants to reduce the spread of HIV/AIDS within his community. Which of the following activities will help him to minimize the spread of the disease? (EGSECE, 2005 E.C)
A. Creating awareness within the community to take vaccination.
B. Mobilizing the community to keep personal hygiene.
C. Mobilizing the community to reduce environmental population
D. Producing posters and leaflets that create awareness about the disease.

Explanation:
- The best way to minimize the spread of HIV/AIDS in a society is creating awareness through education, art, producing posters and leaflets are some of the ways.

Answer: D

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68. List out the diseases

65. Tuberculosis which has once seemed to decrease its effects becomes very dangerous now. What is the most probable reason for this? (EGSECE, 2005 E.C)

A. The spreading of resistant strains of HIV
B. The manifestation and intensification of HIV/AIDS.
C. Pollution of the environment by chemicals
D. Climatic changes that had intensified desertification.

Explanation: **Tuberculosis** is one of the opportunistic infections which attack easily people with weak immunity or resistance. Therefore, since HIV weakens the human immune system, the intensification of HIV/AIDS also increases the spread of tuberculosis.

Answer: B

66. List out examples for – bacterial diseases
– Viral disease – Parasitic worm disease
– protozoan disease

Explanation:

Pathogen	Examples
Viral disease	Small pox, HIV/AIDS, common cold, influenza, etc.
Bacterial diseases	Tuberculosis, pneumonia, Typoid, Gonorrhea, syphilis, etc.
Protozoan diseases	Amoebic dysentery, malaria, sleeping sickness.
Parasitic worm disease	Tapeworm (Tanea saginata, Tenea solum), Ascaris, Hook worm, bilarzia, etc.

67. Describe the advantages of traditional medicines over modern medicines in Ethiopia.

Explanation:

The advantages of using traditional medicines are:

- They are highly available especially in developing countries like Ethiopia.
- People have confidence in them.
- They have holistic effect on treating the patient.
- They are known in the society for many generations.

68. List out the groups of microorganisms that cause the following diseases.

- Tuberculosis
- Typhoid
- Cancroids
- Gonorrhea
- Syphilis
- AIDS

Explanation:

Disease	Cause
Tuberculosis	*Mycobacterium tuberculosis*
Chancroid	*Haemophillus ducreyi*
Syphilis	*Treponema palladium*
Typhoid	*Salmonella typhi*
Gonorrhea	*Neisseria gonorrhea*
AIDS	HIV

4.3. HIV and AIDS

The medical term for AIDS is acquired Immune Deficiency syndrome. This is because it weakens or destroys the immunes system.

AIDS is caused by HIV (Human Immune Virus). Thus, since the virus destroys the immune system it makes the body to suffer with other opportunistic infections like tuberculosis.

> **Remember**: Tuberculosis attacks the body which is in a weak immune state.

HIV can be transmitted through:

- Sexual intercourse which is unprotected.
- From mother to child (baby) in the womb during birth or during breast Feeding.
- Infected blood;
  - By sharing needles during the injection of illegal drugs.
  - When using knives for female genital mutilation.
  - During transfusion of infected blood.

HIV and AIDS in Ethiopia

Ethiopia is among one of the sub Saharan African countries. Thus, HIV/AIDS is a huge problem in Ethiopia.

Globally around 33 million people are infected by HIV/AIDS, in which about 90% of the cases are in sub-Saharan African countries including Ethiopia. The highest rate of infection is on younger people who are in between 15 to 24 years old. In Ethiopia people between the ages of 15 – 49 are highly affected. Thus, HIV/AIDS has a huge impact on the economy of our nation, since it largely affects the working age group.

The prevalence rate of HIV/AIDS is higher in cities than rural areas.

HIV and Immune

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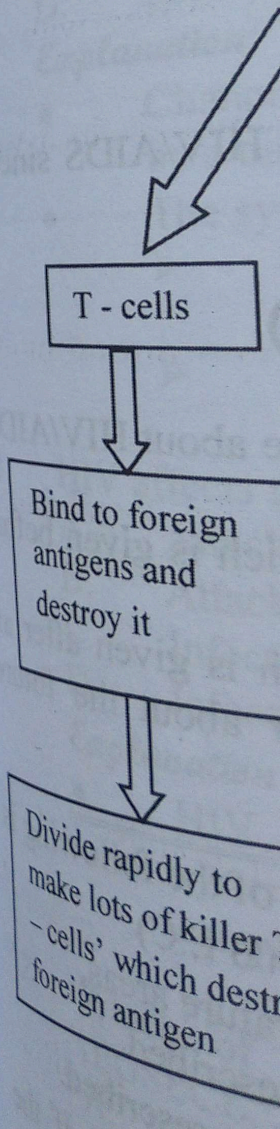

HIV and Immune System

HIV weakens the immune system by destroying T – cell which bind to the antigens on the invading microorganisms and destroy it.
Normally our body has two types of cells that help our immune system. These are **T-cell** and **B – cells.**

- ✓ **T – Cells** are cells that recognize and attach to a cell carrying a foreign antigen and destroy it.
- ✓ **B – Cells** are cells that recognize and attach to a foreign antigen, reproducing and forming many identical copies of themselves which all secrete antibodies to destroy the antigen.

Our body needs the two types of cells for the immune system to work properly

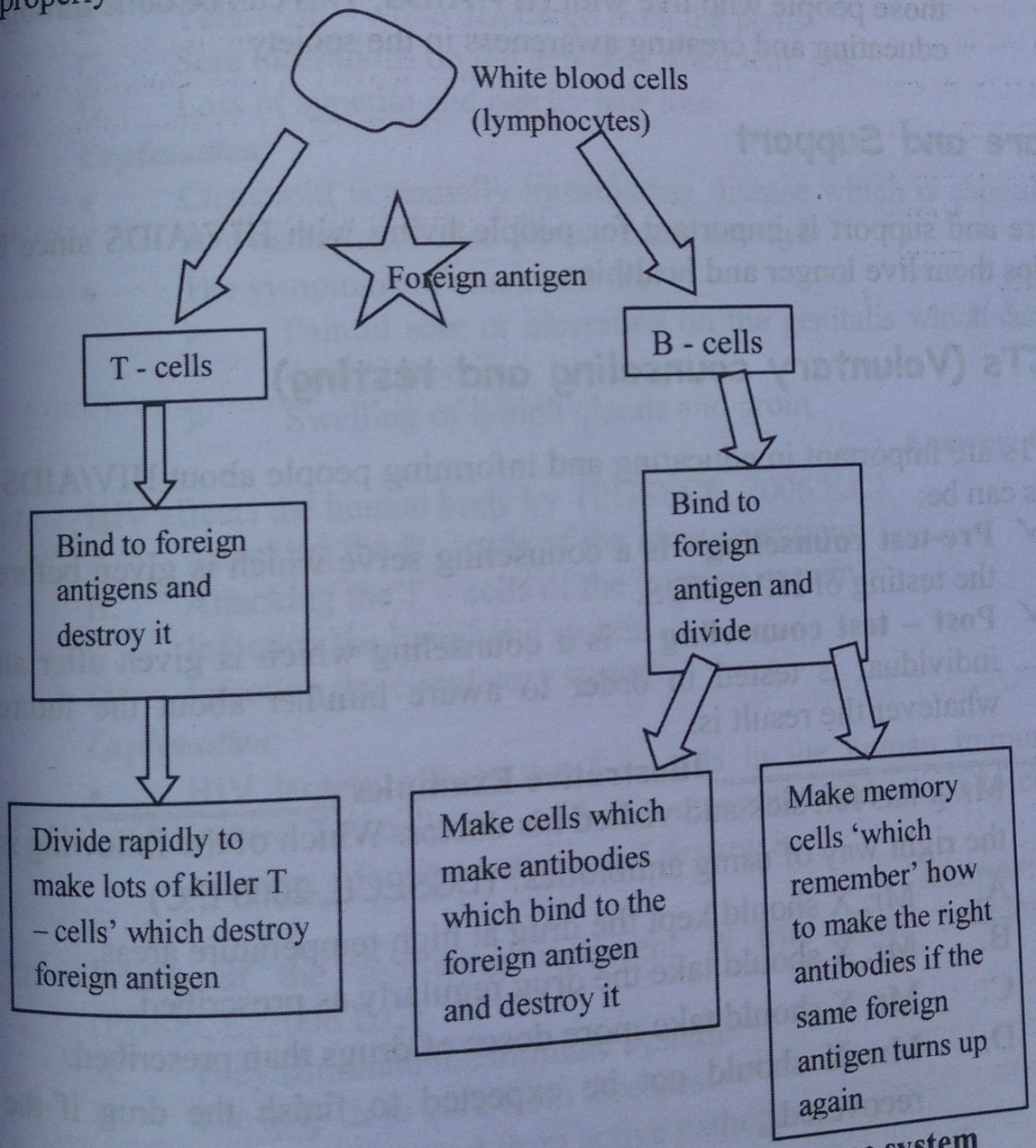

Fig. 4.6: The T – cells and B – cells of the immune system

Stigma and discrimination

Stigma is a mark of disgrace on people with HIV/AIDS, while discrimination is an act of neglecting someone from the group or others. Thus, stigma and discrimination are the most serious cases that affect people living with HIV/AIDS. This may include:

- People donot allow to give them a job.
- People donot allow them to share rooms, dinning table.
- People may neglect their children at school and other places. However, all these are improper and unethical. Thus, we have to fight stigma and discrimination. Instead we have to care and love those people who live with HIV/AIDS. This can be done through educating and creating awareness in the society.

Care and Support

Care and support is important for people living with HIV/AIDS since it helps them live longer and healthier.

VCTs (Voluntary counseling and testing)

VCTs are important in educating and informing people about HIV/AIDS. This can be:

- ✓ **Pre-test counseling** - is a counseling serve which is given before the testing of HIV/AIDS.
- ✓ **Post – test counseling** – is a counseling which is given after an individual is tested in order to aware him/her about the future whatever the result is.

Illustrative Examples

69. Mr. x has felt sick and visited his doctor. Which of the following is the right way of using antibiotics? (EGSECE, 2006 E.C)
    A. Mr. X should kept the drug at high temperature areas.
    B. Mr. X should take the drug regularly as prescribed.
    C. Mr. X should take more doses of drugs than prescribed.
    D. Mr. X should not be expected to finish the drug if the recovered,

Explanation
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70. Which of
2006 E.C)
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71. HIV aff
A.
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72. Which
(EGS
A.
B.

Explanation: When taking medicines including antibiotics, the following precautions should be considered:

- Do not take more than prescribed.
- Do not take less than prescribed.
- Make sure you finish all the medicine as prescribed.
- Do not use a drug which is not prescribed to you.
- Carefully observe the expiry date of the drug before use.

Answer: B

70. Which of the following is a symptom of chancroid? (EGSECE, 2006 E.C)
   A. Burning sensation while urinating.
   B. Yellowish – white discharge from the penis
   C. Sore ulcerations on the genitalis filled with pus.
   D. Loss of appetite and patchy hair loss.

Explanation:

- Chancroid is sexually transmitting disease which is caused by a bacterium called ***Hemophilus ducreyi.***
- The symptoms of chancroid are:
  - Painful sore or ulceration on the genitalis which can easily bleed.
  - Swelling of lymph glands and groin.

Answer: C

71. HIV affects the human body by (EGSECE, 2006 E.C)
   A. Attacking the B – cells of the immune system.
   B. Attacking the T – cells of the immune system.
   C. Infecting the lymphatic system.
   D. Infecting the circulatory system.

Explanation:

- HIV largely affects the T – cells in the human immune system T – cells are cells that recognize and attach to cell carrying a foreign antigen and destroy it.

Answer: C

72. Which of the following statement is true about vaccines? (EGSECE, 2006 E.C)
   A. They stimulate the immune system.
   B. The help us to cure from diseases.
   C. They are produced from active pathogens.
   D. They are given only in needles.

Explanation: Vaccines work by stimulating the immune system of the body. Thus, it helps the body to develop the antibodies to a disease. It involves introducing weakened forms of pathogens in the body.

Answer: A

73. Living organisms that can be seen only with the aid of microscope are called: (EGSECE, 2006 E.C)

A. Yeast
B. Bacteria
C. microorganisms
D. microorganisms

Explanation:
Microorganisms are very small organisms that cannot be seen by our naked eye. Thus, they can be seen only with the aid of microscope.

Answer: D

74. Which of the following statement is true about microorganisms? (EGSECE, 2006 E.C)

A. All microorganisms are important sources of food.
B. Some microorganisms are important in nutrient decomposition.
C. Many microorganisms are disease causing.
D. All microorganisms are poisonous.

Explanation:

- Some microorganisms are important sources of food.
- Some microorganisms are important in nutrient decomposition.
- Some microorganisms are disease causing.
- Some microorganisms are poisonous.

Answer: B

75. Tape worm is one of human parasite. It is: (EGSECE, 2006 E.C)

A. Caused by the bacterial infection of intestine.
B. Prevented by avoiding contact with mosquito.
C. Prevented by anti – inflammatory drugs
D. Caused by parasitic plasmodium.

Explanation: Tape worm is the human intestinal parasite that is:

A. Caused by the bacterial infection of intestine.
B. Prevented by avoiding contact with mosquito.
C. Prevented by anti – inflammatory drugs.
D. Caused by parasitic plasmodium.

Explanation: Tape worm is the human intestinal parasite that is controlled and prevented by:

- Proper disposal of feaces.

- Proper inspection of meat.
- Avoiding eating raw and poorly cooked meat.
- Using anti – worm inflammatory drug.

Answer: C

76. Which of the following statements is true about tuberculosis? (EGSECE, 2006 E.C)
   A. It is caused by viruses and bacteria.
   B. It affects the heart and lung.
   C. It has a symptom of abdominal cramps.
   D. It has a symptom of night sweat.

Explanation: Tuberculosis

- is caused by a bacterium ***mycobacterium tuberculosis.***
- Its symptoms include:
  - mild fever
  - weight loss
  - night sweat
  - persistent cough
  - fatigue
- It largely affects the lung, but also some other body parts can be affected.

Answer: D

Supplementary Exercise

1. Which of the following is incorrect about bacteria's?
   A. They are unicellular
   B. They are prokaryotic
   C. they reproduce sexually
   D. they have cell wall
2. All of the following are useful aspects of bacteria's, except:
   A. Making of vaccines
   B. Increasing soil fertility
   C. Production of alcohol
   D. tanning of lather
3. All of the following are true about viruses, except.
   A. They are obligate parasites
   B. They have genetic material
   C. They cannot reproduce outside the living cell
   D. They can synthesize their own food
4. Which of the following scientists are accredited for the germ theory of disease?
   A. Louis Pasteur and Robert Hooke
   B. A.V. Leeuwenhoek and Theodore Schwann
   C. Louis Pasteur and Robert Koch
   D. Robert Hooke and Aristotle
5. The first person who demonstrated, certain diseases are caused by bacteria's is:
   A. A.V. Leeuwenhoek
   B. Robert Koch
   C. Lazaro Spalanzani
   D. Aristotle

6. The microbiological method which involves killing of all the microbes on any object is:
   A. Sterilization C. disinfection
   B. Pasteurization D. antiseptics
7. The most effective sterilization technique which is used to sterilize scientific equipments is:
   A. Ultra high temperature C. autoclave sterilization
   B. Dry heat sterilization D. irradiation
8. A microbiological method which involves use of chemicals to kitchen equipment, toilet, sinks is:
   A. Antiseptics C. disinfectant
   B. Irradiation D. pasteurization
9. A nutrient medium which is extracted from red algae is:
   A. agar B. broth C. culture D. All are possible
10. Graham +ve bacteria's
   A. Decolorized and loose their primary stain
   B. Retain their primary stain
   C. Have no cell wall
   D. Have purple color
11. When a baby is born with antibodies from the mother. This is an example of
   A. Natural passive immunity
   B. Artificial active immunity
   C. Natural active immunity
   D. artificial passive immunity
12. Special protein produced by the white blood cells to fight infectious microorganisms are:
   A. Pathogens C. antibodies
   B. Antigens D. vectors
13. Which of the following cannot be considered as inherited immunity?
   A. HCl in the stomach C. lymphocytes
   B. Phagocytes D. antibiotics
14. The process of introducing dead or weakened forms of pathogens in to the body, in order to stimulate it to produce its own antibodies is:
   A. Natural immunity C. passive immunity
   B. Active immunity D. inherited immunity

15. Tuberculosis
   A. Good
   B. Avoid
   C. good
   D. pasteu
16. Eating or dri
   is NOT the c
   A. syphil
   B. chole
17. Which of th
   A. HIV/
   B. Tube
18. A segments
   is:
   A. Scole
   B. Suck
19. Among the
   do have hig
   A. Gon
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20. Sexually tr
   A. Avo
   B. Bei
   C. Earl
   D. Avo
21. The advan
   A. The
   B. The
   C. the
   D. the
22. Which of
   from dise
   A. Mo
   B. Iso
23. Which of
   A. Ty
   B. Ho

15. Tuberculosis cannot be prevented by:
A. Good ventilation
B. Avoid eating raw meat
C. good personal hygiene
D. pasteurization of milk

16. Eating or drinking contaminated food and water with feacal waste is NOT the cause for:
A. syphilis
B. cholera
C. gastrointestis
D. Typhoid

17. Which of the following is a vector borne disease
A. HIV/AIDS
B. Tuberculosis
C. Tape worm
D. malaria

18. A segments of tapeworm which contains male and female organs is:
A. Scolex
B. Sucker
C. cysticerci
D. progolottids

19. Among the following sexually transmitted diseases, the one which do have high risk of HIV transmission is :
A. Gonorrhea
B. Syphilis
C. chancroid
D. schistosomia

20. Sexually transmitted diseases cannot be controlled by
A. Avoiding unprotected sexual intercourse
B. Being faithful to one sexual partner
C. Early treatment
D. Avoiding droplet infection

21. The advantage of modern medicines over traditional medicines is:
A. Their holistic effect
B. They have limited doze
C. they are relatively cheep
D. their availability

22. Which of the following is the best method of preserving oneself from diseases?
A. Modern medicine
B. Isolation
C. personal hygiene
D. Traditional medicine

23. Which of the following types of diseases is caused by virus?
A. Typoid
B. Hookworm
C. Tuberculosis
D. Rabies

24. Which of the following is transmitted by a vector?
    A. Tuberclosis
    B. Malaria
    C. Mumps
    D. Aids
25. The first scientist who discovered disinfectants was:
    A. John Needham
    B. Fransco Redi
    C. Joseph Lister
    D. Robert Koch
26. Suppose you want to preserve your milk by eliminating disease causing organisms. Which of the following methods would you use?
    A. Boiling the milk in water 120^0C under high pressure
    B. Heating the milk for 2 hours at 160^0C.
    C. Heating the milk for 30 minutes at 63^0C.
    D. Exposing the milk to UV – radiation.
27. Typhoid fever can be transmitted through:
    A. Cough and sneezing
    B. Direct contact
    C. infected water and food
    D. insected bite

Matching

28. Autoclave sterilization	A. 63°C for 30 minutes
29. Ultra high temperature	B. 171°C for 1 hour
30. Pasteurization	C. 121°C for 15 – 45 minutes
31. Dry heat sterilization	D. 135°C – 150°C for 2 – 6second

Fill in the blank spaces

32. The study of microorganisms is called _____.
33. The transfer of microorganism in to a nutrient medium is called _____.
34. _____ is a technique of boiling a food which is followed by rapid cooling.
35. _____ is the theory which states that certain non – living things can be directly changed in to a living thing.
36. Disease causing microorganisms are called _____.
37. _____ is a solid nutrient medium which is made of red algae.
38. Proteins that are foreign to the body are _____.
39. _____ are disinfectants that are applied to the skin.

40. Drugs th
41. An STD
42. microorg

1. A 2.

Part I: 1. A 2
Part II: 1. D

Part I: 1. A 2
Part II: 1. C

1. D 2. A
2.

1. C
6. C
11. A
16. A
21. B
26. C

Matching

28. C

Fill in the B

32. Micr
35. Theo
36. Patho
39. Antis
42. Nutr

40. Drugs that are used to treat bacterial diseases are called _____.
41. An STD which increases the risk of HIV/AIDS infection is _____.
42. _____ is the collection of nutrients that are used to grow microorganisms.

Answers to Post Test 1

1. A 2. C 3. B 4. D 5. C 6. C 7. D

Answers to Post Test 2

Part I: 1. A 2. A 3. B 4. B 5. C 6. B
Part II: 1. D 2. E 3. A 4. B 5. E

Answers to Post Test 3

Part I: 1. A 2. A 3. B 4. C 5. A
Part II: 1. C 2. D 3. A 4. B 5.B

Answers to Post Test 4

1. D 2. A 3. E 4. F 5. B 6. C
2.

Answer to Supplementary Exercises

1.	C	2.	C	3.	D	4.	C	5.	B
6.	C	7.	B	8.	C	9.	A	10.	B
11.	A	12.	C	13.	D	14.	B	15.	B
16.	A	17.	D	18.	D	19.	C	20.	D
21.	B	22.	C	23.	D	24.	B	25.	C
26.	C	27.	C						

Matching

28. C 29. D 30. B 31. A

Fill in the Blank

32. Microbiology 33. Inoculation 34. Pasteurization
35. Theory of spontaneous generation
36. Pathogens 37. Agar 38. Antigens
39. Antiseptics 40. Antibiotics 41. Chancroid
42. Nutrient Medium

5. CLASSIFICATION

5.1. Principle of Classification

The world in which we are living is a home for variety of living things. It is estimated that there are millions of species of organisms among which only 1.5 million species are identified and studied. Therefore, it's important to classify organisms:

- To study the relationship between organisms in order to understand how they are originated.
- To place them in groups on the basis of their similarities and differences in order to avoid chaos and confusions.
- To simplify their study by giving them precise names.

The Classification System

The detail aspects about the classification of organisms, it is better to get familiar with the following important terminologies.

- **Species** – the group of organisms that can breed successfully with one another to produce fertile offspring.
- **Classification** – is the system of grouping organisms based on their similarities and differences.
- **Taxonomy** – it is the branch of biology which is concerned with classification of organisms. (Greek, **Taxis – to arrange, nomos - law**)

Different methods of classification were introduced by different biologists at different times.

✓ **Aristotle** – is a Greek philosopher who attempted to classify organisms for the first time in the 400 B.C.

- First he grouped living things in to two groups – **plants** and **animals.**

- Again he classified animals in to three groups;
  - Animals that can fly
  - Animals that live in water
  - Animals that live on the land.
- Plants based on their size and pattern of growth;
  - Tall plants
  - Medium plants
  - Small plants

The modern classification method is introduced by a Swedish botanist **Carl Linnaeus**. He classified all plants and animals based on their similarities and differences. This method followed up the evolutionary relationship between organisms. Thus, it is called the **natural classification method**.

The modern classification method involves the following categorization system.

Species→ Genus → Family → order → class → phylum → kingdom

- Species refers to the group of organisms that can interbreed successfully and produce fertile off spring.
- Organisms of similar or related species are grouped in to a **genus.**
- Organisms of similar or related genus are grouped in to a **Family.**
- Organisms of similar or related family are grouped in to an **order.**
- Organisms of similar or related order are grouped in to a **class.**
- Organisms of similar or related class are grouped in to a **phylum.**
- Organisms of similar or related phyla are grouped in to a **kingdom.**

Naming of Livings

Carl Linnaeus devised a **scientific name** for organisms which consists of two names. The first name is genus name which is followed by species name. This naming system is called **binomial nomenclature**

- For example, the scientific name for human being is *Homo sapiens*
- The scientific naming of organisms follows the following rules,
  - The first name is a genus name in which it starts by capital letter.

- The second name is species name which is all written in small letters.
- The two names are underlined or written in italics.

➢ **Scientific names** are needed in study of organisms, since it helps all the taxonomists to have a common understanding by replacing the common names used by different languages.

➢ **Common names** are names given to organisms by different languages.

- ***For example***: Dog – in English, ውሻ - in Amharic, Saree – in oromiffaa. However, since its difficult to use this names by scientists at different corners of the world, using **scientific names** is preferable.

Table 5.1: Examples of scientific names of some organisms

Common name	Scientific name	Genus name	Species name
Domestic cat	*Felis domesticus*	*Felis*	*domestics*
Dog	*Canis familiaris*	*Canis*	*familiaris*
Human being	*Homo sapiens*	*Homo*	*Sapiens*
Kosso tree	*Hagenia abysinica*	*Hagenia*	*abysinica*
Lion	*Panthera leo*	*Panthera*	*leo*
Leopard	*Panthera pardus*	*Panthera*	*Pardus*
Maize	*Zea mays*	*Zea*	*mays*

Table 5.2: The Hierarchy of groups

	Human	Domestic cat	Leopard	Teff	Honey bee	Mushroom
Kingdom	Animalia	Animalia	Animalia	plantae	Animalia	fungi
Phylum	Chordata	Chordata	Chordata	Angiosper mophyta	Arthropod a	Basidomeyc ota
Class	Mammalia	Mammalia	Mammali a	Liliophsid a	Insecta	Basidomyce tes
Order	Primates	Carnivore	Carnivore	Cyperales	Hymen optera	Agariceae
Family	Hominidae	Felidae	Felidae	Poaceae	Apidae	Agariceae
Genus	Homo	Felis	Phanthera	Eragrostis	Apis	Agaris
Species	Sapiens	Domesticus	Pardus	Teff	Melifera	Campstris

Illustrative Example

1. Scientific name is important for scientists because:
   A. Each name has two parts
   B. It is written in Latin
   C. Scientists of different countries use different scientific names.
   D. They use only one scientific name for one species.

Explanation: Scientific name is important because one species of an organism has only one scientific name which helps scientists of different countries to have common understanding

Answer: D

2. The scientific name for modern man is:
   A. *Homo erectus* C. *Homo neanderthalis*
   B. *Homo sapiens* D. *Austalopithecus*

Explanation: ***Autralopiticus*** and ***Homo neandertals*** lower primates which have a relationship with fossil of man.

- **Homo erectus** evolved to modern man (*Homo sapiens)*
- ***Homo sapiens*** is a modern man

Answer: B

3. A group of organisms consisting of one or more genera is a
   A. Family C. genus
   B. Species D. class

Explanation: Genus its plural 'Genera'

- Similar **species** are grouped into a **genus.**
- Similar **genera** are grouped into a **family.**
- Similar **families** are grouped into a **order.**
- Similar **orders** are grouped into a **class.**
- Similar **classes** are grouped into a **phylum.**
- Similar **phyla** are grouped into a **kingdom.**

Answer: A

4. What statement can be made about members of the same order? (EGSECE 1997 E.C)
   A. They all belong to the same genus.
   B. They all belong to the same class.
   C. They can all interbreed successfully.
   D. They all belong to the same family.

Explanation:
Similar orders are grouped in to the same class. (Look at example number 3)

Answer: B

5. Which of the following scientific name is correctly written? (EGSECE, 2006 E.C)
   A. *Eragrostis teff*
   B. *Musca domesticus*
   C. *Homo sapiens*
   D. *canis Familiaris*

Explanation: When writing a scientific name the following rules should be followed:

- The first name is genus name which starts with a capital letter.
- The second name is species name which is all written in small letters.
- The two names are written in italic font in a computer print or underlined if hand written.

Answer: A

6. Which of the following is correct about the scientific name of an organism?
   A. It is the same for scientists all over the world.
   B. It varies according to the native language of scientists.
   C. It may have more than one genus name.
   D. It may refer to more than one species.

Explanation: A scientific name used for one species is the same all over the world and one species has only one scientific name.

Answer: A

7. Which of the following shows an orderly sequence when going from broad groups to narrower ones? (EGSECE 1975 E.C)
   A. Phylum →Family → class → order
   B. class → phylum → order → family
   C. Family → order → class → phylum
   D. Phylum→ class → order → family
   E. Family → order → phylum → class

Explanation: The correct hierarchy from the broader to the simpler is

Kingdom → phylum → class → order → family → Genus →species

Answer: D

8. Which of the following is true about the scientific name *Apis melifera*? (EGSECE 2004 E.C)
   A. *Apis* is the class name
   B. *melifera* is the species name
   C. *Apis* is the family name
   D. *melifera* is the order name

Explanation
and the sec
Apis melif
name
Which of
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B. Phy
C. Phy
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5.

Explanation: In a scientific name, the first name is a genus name and the second name is species name. Thus in a scientific name ***Apis melifera***, ***Apis*** is a **genus** name and ***melifera*** is ***species*** name

Answer: B

9. Which of the following taxonomic categories of organisms is in the correct sequence? (EGSECE, 2005 E.C)
   A. Phylum → family →class → genus → order →species
   B. Phylum → order → family → class → genus →species
   C. Phylum → class → family → order → genus →species
   D. Phylum → class → order → family → genus →species

Explanation: The correct order of taxonomic category from broader to narrower is: Kingdom → Phylum → class → order → family→ genus →species

Answer: D

10. The ability to interbreed in nature and produce fertile offspring is the characteristics of members of a single (EGSECE 1997 E.C)
    A. Order
    B. Species
    C. phylum
    D. family

Explanation: organisms of the same species have an ability to breed successfully to bring fertile offspring. Species is the lowest taxonomic hierarchy

Answer: B

11. Which of the following is correct about organisms as we move through the biological hierarchy from the kingdom to species level?
    A. They are less and less related to each other.
    B. They vary more and more.
    C. They always are members of the same family
    D. They become more similar in appearance

Explanation: As we go through a taxonomic hierarchy (from kingdom to species) organisms become more and more similar.

Answer: D

Item number 18 is based on the following table which shows the five kingdoms and their characteristics

No.	Kingdom	Characteristics
1.	Monera	Eukaryotic, single celled
2.	Protista	Prokaryotic single celled
3.	Fungi	Eukaryotic multi cellular
4.	Plant	Prokaryotic multi cellular
5.	Animal	prokaryotic multi cellular

12. Which kingdom is correctly paired with its characteristics? (EGSECE 2004 E.C)
A. 1,3,4 and 5 C. 1
B. 1,2,3,4 and 5 D. 3

Explanation: the correct pair is:

Kingdom	Characteristics
Monera .. . . →	Prokaryotic, single celled
Protista .. . . →	Eukaryotic, single celled
Fungi . . .. . →	Eukaryotic, multi cellular and some single celled
Plantae .. . . →	Eukaryotic, multi cellular
Animalia .... →	Eukaryotic, multi cellular

Answer: D

13. Which of the following is the largest taxonomic category? (EGSECE 2004 E.C)
A. Class B. Genus C. family D. order

Explanation: The correct order according to their complexity (from complex to simple) is:
Class → order → family → genus

Answer: A

12. Why is it important to classify organisms?

Explanation:
The natural method of classification which is mainly based on the evolutionary relationship is important because:

- To study the relationship between organisms.
- To place them in groups based on their similarities and differences.
- To simplify their study

13. Write the correct hierarchy of classification of organisms from simpler to complex categorization.

Explanation:

- The most complex hierarchy in the classification of organisms is **kingdom**, while the simplest scheme in classification is **species**. Thus, arrangement of organisms from simpler to complex is:

Species → genus → family → order → class → phylum → kingdom

14. Why are scientific names more acceptable than the common names in the study of classification of organisms?

Explanation: Common names are names which are given to an organism by different languages. Thus, scientific names which are names given to organisms by scientists and it helps scientists all over the world to have common understanding.

Part I: Choose

1. Which
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Post Test 1

Part I: Choose the best answer from the given alternatives.

1. Which of the following is not listed under the five kingdom system of classification?
   A. Kingdom Fungi
   B. Kingdom Plantae
   C. Kingdom Algae
   D. Kingdom monera
2. The correct order of hierarchy of classification which one has the smallest taxonomic category:
   A. Class B. Genus C. Phylum C. Order
3. Which one of the following is the correct writing of a scientific name?
   A. *Canis Familiaris*
   B. *canis familiaris*
   C. Canis familiaris
   D. *Canis familiaris*
4. In a scientific name *Homo sapiens*; *sapiens* belong to:
   A. Kingdom name
   B. Class name
   C. Species name
   D. Genus name
5. A kingdom having prokaryotic single celled organisms belong to:
   A. Monera
   B. Plantae
   C. Animalia
   D. Monera

5.2. The Five Kingdom System

Even though different schemes of classification were introduced at different times, the most widely accepted one is the '**five kingdom system**' which is introduced by an American biologist. **R.H. Whittaker** by the year 1969.

The five kingdom system consists of the following major kingdoms.

1. **Kingdom monera** – consists of bacteria and blue green algae.
2. **Kingdom protista** – consists of eukaryotic single celled organisms such as Euglena, Diatoms, paramecium, Dinoflagellate, etc.
3. **Kingdom Fungi** – consists of eukaryotic organisms such as yeast moulds, mushrooms.
4. **Kingdom plantae** – consists of all multicellular, eukaryotic and autotrophic plants.
5. **Kingdom animalia** – consist of all multicellular, eukaryotic and heterotrophic animals.

1. Kingdom monera

Moneras are the most plentiful organisms on the earth.

- They live almost everywhere; water, wet surfaces, rocks, soil, desert, parasites.
- The representative groups are **eubacteria** (true bacteria) and **cyanobacteria** (Blue – green algae).

Some important characteristics of moneras are:

- They are unicellular and prokaryotic.
- They have either autotrophic or heterotrophic modes of nutrition.
- Most of them have cell wall.
- They reproduce asexually by Fragmentation and binary Fusion.
- Some of them are harmful and cause disease. For example:
  - *Mycobacterium tuberculosis* causes tuberculosis.
  - ***Hemophilus decreyi*** causes chancroid etc.
- Others are useful to humans.

2. Kingdom protista

Protists consist of usually unicellular, eukaryotic organisms. However some biologists classify multicellular algae such as brown algae in to this kingdom.

Representative groups of this kingdom are **subkingdom protozoa** and **sub kingdom algae.**

Subkingdom Protozoa

- They are microscopic and have no chloroplasts.
- They live as a parasite or free living.
- Have locomotory structures.
  - Amoeba – pseudopodia
  - Paramecium - cilia
- Consists of the following phylas:
  - **Phylum sarcomastigophora** – consists of amoebas and Flagellates like Giardia and Trypanosomes.
  - **Phylum ciliophora** – consists of ciliates like paramecium.

Phylum
parasite

Fig. 5

Sub Kingdom

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live on the
Their bod
undifferent
Sub kingdo
a) Phy

b) P

c) E

- **Phylum Apicomplexa** – consists of well known human parasite called **Plasmodium**.

Fig. 5.1: paramecium

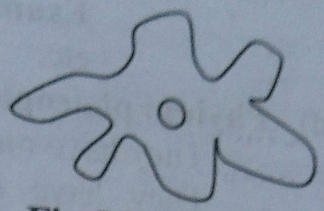


Fig.5.2. Amoeba

Sub Kingdom Algae

- They are photosynthetic and eukaryotic
- They usually live in aquatic habitat and some of them adapted to live on the land.
- Their bodies lack true stem, root and leaves. They have undifferentiated body called **thallus**.
- Sub kingdom algae consists of the following phyla:

a) **Phylum Dinoflagella**

- They have yellow – brown pigment
- They have flagella for locomotion. Examples: cryptomonas, ceratium
- They cause bioluminescence in the seas and oceans when they produce greenish light.

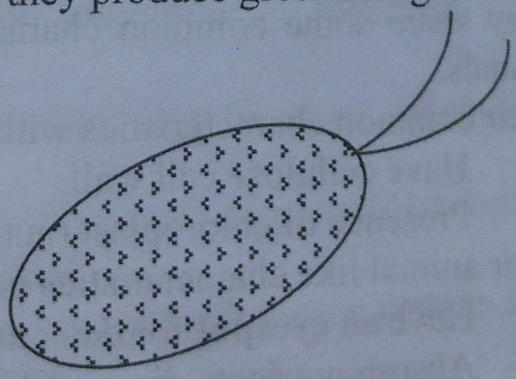


Fig. 5.3.: Cryptomonas

b) **Phylum Bacillariophyta (Diatoms)**

- They have yellowish – brown pigment
- They are sources of food for aquatic animals

Example: Diatoms

c) **Phylum chlorophyta (green algae)**

- They are unicellular, non-motile.

- They contain dominant photosynthetic pigment called **chlorophyll-a** and **chlorophyll-b.**
- They serve as a base for aquatic food chain.
- **Examples**: chlorella, chlamydomonas, spirogyra, etc.

d) Phylum phaeophyta ("brown algae")

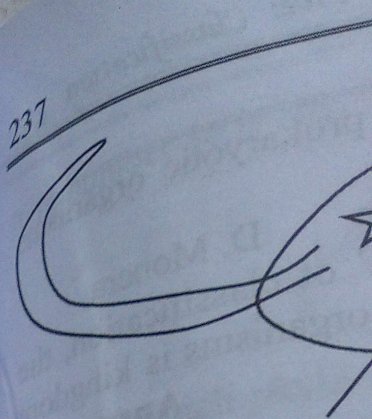
- They have multicellular body organization.
- They have dominant photosynthetic pigment which is brown and called **Fucoxantin**
- Used as thickening agents in ice cream and cosmetic industries. They are also rich in their iodine content which can be used for human consumption.
  Examples: Fucus, Laminaria

e) Phylum Rhodophyta (red algae)

- Have multicellualr body organization with filamentous or feathery bodies. Some are unicellular.
- They contain additional red pigmented substance (**phycoerythrin**) in addition to chlorophyll.
- They are the main constituents of **agar** which as a solid nutrient medium. They can also be used as constituents in making laxatives and tooth paste.

f) Phylum euglenophyta

- They are unicellular Flagellates.
- They share some common characteristics with plants and animals.
- Their common characteristics with plants are:
  - Have cellulose cell wall.
  - Presence of chloroplast (authotrophic).
- Their animal like characteristics include:
  - Have an **eyespot** for detecting the direction of light.
  - Absorb nutrients dissolved in water (heterotrophic).
  - Have flagella for locomotion
  - Serve as source of food for some aquatic organisms and also serve as indicators of pollutants.



Eyespo

3. Kingdom Fung

- They have he
  chlorophyll.
- They can be pa
- Digestion takes
- They have rig
  containing poly
- They store carb
- **Parasitic Fung**
  Example: *C*

- **Saprophytic**
  decomposers
  ecosystem.
- **Mutialist fu**
  organisms.
  Example: **Lic**

Mycorrhizae

- Kingdom Fun
  - **Phylu**
    mucor
  - **Phylu**
  - **Phylu**

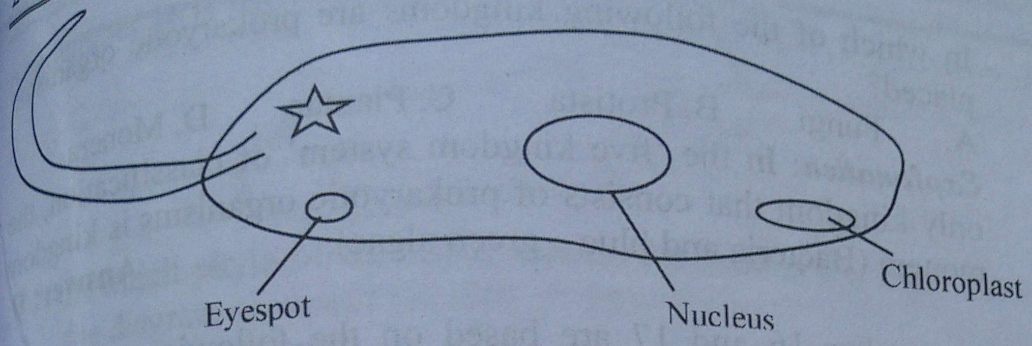


Fig. 5.4: Euglena

3. **Kingdom Fungi**

- They have heterotrophic mode of nutrition since they lack chlorophyll.
- They can be parasitic, saphrophytic or mutualists.
- Digestion takes place outside their body.
- They have rigid cell wall which is made up of a nitrogen containing polysaccharide called **chitin**.
- They store carbohydrate in the form of glycogen,
- **Parasitic Fungi** – feed on living tissues plants and animals.
- **Example**: ***Candida albicans – thrush fungi, Tenea pedis*** – Athletes foot fugi, Mildew- plant parasites.
- **Saprophytic Fungi** – They feed on deed materials. They act as decomposers which play role in recycling of matter in an ecosystem.
- **Mutialist fungi** – they live in close association with other organisms.
  Example: **Lichen** – a mutualistic association between **algae** and **Fungi**.
  Mycorrhizae – an association between **fungus** and **roots of the plant.**
- Kingdom Fungi consists of three major phylas:
  - **Phylum zygomycota** – includes Rhizopus, mould and mucor.
  - **Phylum Ascomycota** – mushrom, yeast, mildews.
  - **Phylum Basidmycota** – Mushrom, Agaricus campestris.

Illustrative Examples

15. In which of the following kingdoms are prokaryotic organisms placed?
A. Fungi B. Protista C. Plantae D. Monera

Explanation: In the 'five kingdom system' of classification, the only kingdom that consists of prokaryotic organisms is kingdom monera (Bacteria and blue – green algae). **Answer: D**

Question number 16 and 17 are based on the following organisms: (ESLCE – 1988 E.C)

1. Spirogyra 3. Amoeba 5. Yeast 7. Scenedesmus
2. Paramecium 4. Rhizopus 6. Penicillum 8. diatoms

16. Organisms that are normally placed in the group fungi are:
A. 1,3,5 C. 2,6,7 E. 4,5,6
B. 3,4,8 D. 1,7,8 **Answer: E**

17. Organisms that are normally classified with the group Algae are:
A. 1,7,8 C. 1,2,3 E. 2,3,5
B. 4,5,6 D. 1,4,7

Explanation: Among the list given above some organisms are grouped under fungi, while others under group algae and protozoa shown in table below.

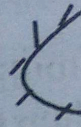
Algae	Fungi	Protozoa
Spirogra	Yeast	Ameoba
Diatoms	Yeast	Paramecium
Scenedesmus	Rhizopus	

Answer: A

18. When you compare kingdom protista in five kingdom systems of classification, protista does NOT include:
A. Amoeba C. Algae
B. Fungi D. Paramocium

Explanation:
Kingdom protista includes all protozoans and algaes. Thus, fungi is included under kingdom fungi, while amoeba and paramecium are protozoans **Answer: C**

Item number **16** is based



19. In which phyla
the diagram gro
A. Ciliopho
B. Sacomas
Explanation:
An organism i
classified unde

20. List out the
introduced by
Explanation:
➢
➢
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21. Why are n
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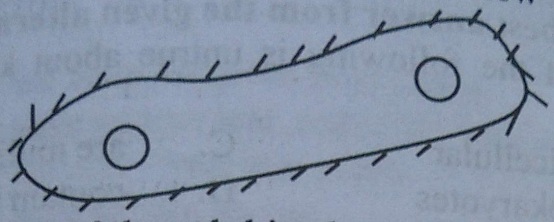
Item number **16** is based on the diagram given below



19. In which phyla of the sub kingdom protozoa is the organism on the diagram grouped? (EGSECE, 2003 E.C)

A. Ciliophora
B. Sacomastigophora
C. rhodophyta
D. Apicomplexa

Explanation:
An organism indicated on the diagram is paramecium which is classified under phylum ciliophora.

Answer: A

20. List out the five kingdom system of classification which is introduced by R.H. whit taker.

Explanation: The five kingdom system consists of:

- Kingdom monera
- Kingdom protista
- Kingdom fungi
- Kingdom platae.
- Kingdom Animalia

21. Why are moneras so different from organisms of other kingdoms?

Explanation:

- Organisms of kingdom monera differ from other kingdom since they are the only prokaryotic groups.

22. List out the three phyla of sub kingdom protozoa.

Explanation: A Sub kingdom protozoon consists of the following phyla:

- Phylum sarcomastigophora.
- Phylum ciliophora
- Phylum apicomplexa.

23. List out the major phylas of kingdom fungi.

Explanation:
Kingdom fungi consist of the following plays. These are:

- Phylum zygomycota – examples: Mould, mucor, Rhizopus.
- Phylum ascomycota – examples: penicillium, yeast, mildew.
- Phylum Basidomycota – examples: Mushrooms.

Post test 2

Part I: Choose the best answer from the given alternatives.

1. Which one of the following is untrue about kingdom monera? They :
   A. Are unicellular
   B. Are prokaryotes
   C. are mulitcellular
   D. contain bacteria
2. Which of the following organisms is categorized under kingdom protista?
   A. Blue green algae
   B. Amoeba
   C. Bacteria
   D. Yeast
3. All of the following are funguses except:
   A. Yeast
   B. Mushroom
   C. Rizopus
   D. Paramacium
4. The locomotory structure if euglena is:
   A. Cilia
   B. Pseudopodia
   C. Falgella
   D. Antennae
5. The only prokaryotic organism are found in the:
   A. Kingdom monera
   B. Kingdom protsta
   C. Kingdom fungi
   D. Kingdom animalia

4. The plant kingdom

The main characteristics of plants are:

- ✓ They have eukaryotic cells.
- ✓ They are multicellular organisms.
- ✓ They contain chlorophyll and carry out photosynthesis.
- ✓ They are predominantly land dwelling.
- ✓ They are autotrophic.
- ✓ Their cell walls are composed of cellulose.
- ✓ They reproduce sexually and asexually.

The plant kingdom consists of the following divisions.

1. Division Bryophyta (mosses and liverworts)
2. Division pteridophyta (ferns)
3. Division lycophyta (club mosses)
4. Division sphenophyta (horse tail)
5. Division Gymnospermae (conifers)
6. Division Angiospermae (flowering plants)

Division Bryophyta (Mosses and liverworts)

- They are simplest green land plants.
- They have no true leaf, root and stem.
- They are non – vascular plants.
- They have simple leaves.
- They have simple root like structures called **rhizoid** which is used for absorption of water.
- **Note:** Vascular plants are those plants that have xylem and phloem tube in their roots, stems and leaves.
- They are commonly found in rainforests and high attitude mountains.

Division Lycophyta (Club mosses)

- They are also small green plants.
- They posses true roots, stem and small scale like leaves.
- Their leaves are densely arranged on branching stem with sporangia.

Division Sphenophyta (Horse tail)

- They are small green plants that have true roots, jointed stems and leaves.
- They have spore producing structures called **strobilus**

Division Pteridophyta (Ferns)

- They have large prominent leaves called **fronds** and horizontal underground stem called rhizome which bears roots.
- They have vascular tissues (xylem and phloem).
- They spores are produced in the fronds and their spores are dispersed by wind.
- They live in dampy shady places and very common in tropical rainforest.
- They reproduce by **alternation of generation**. The **sporophyte** stage is well developed while their gametophyte stage is reduced

Example: Dryopteris filix – mas

- Pterium species exceptionally grow in full sunlight

Division Gymnospermae (Non - flowering plants)

The gymnosperms are higher plants that have well developed root, stem and leaves.

- They have vascular tissues (xylem and phloem tissues)
- They have unprotected ovules by ovary and they from "naked seeds" which are not enclosed in the fruit.

N.B: Ovule is the part of the seed which develops in to a seed and ovary is a part that develops in to a fruit.

- They leaves are mostly needle shaped and ever green.
- Their reproductive structures are found in structures called **cones**. Male cone and female cones are found on different parts of the plant. They are heterosporous (microspores and mega spores).

  Microspores – are male gametophyte.

  Megaspores – are Female gametophyte.
- They have no vessels in xylem, only tracheids.

  Examples of gymnosperms in deginous to Ethiopia are Tid (ጽድ) – *Juniperus procera* and Zigba (ዝግባ) – ***Podocarpus falcutus.***

- The largest groups of gymnosperms are conifers.

Division Angiospermae (Flowering plants)

Angiosperms (flowering plants) are higher plants that posses true stem, leaves, roots and flowers.

- They are the most successful land plants.
- They produce seeds which are covered by Fruits.
- They posses a well developed vascular tissues.
- They have highly developed asexual and sexual reproduction.
- Xylem contains vessels and tracheids.
- Ovules are protected with in ovary. The ovule develops in to seed and ovary develops in to fruit.
- Angiosperms are divided in to two classes. These are **monocots** and **dicots**.

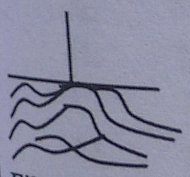
i) **Monocotyledons**
parallel venat
• They are m
• Example:

ii) **Dicotyledons** –
with netted v
• They are herb
beans, etc.

Mono

- The embry cotyledon.
- Leaves hav Venation.

- Steam co bundles in s
- Have Fibro roots.



Fibrous root

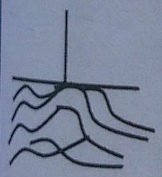
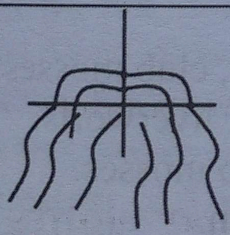
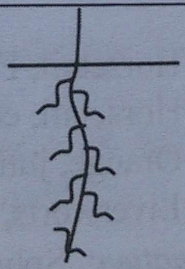
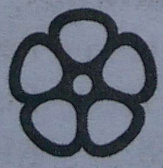
- Have need
- Floral pa multiple o
- Have tru with va absent.

Example

i) **Monocotyledons** - have seeds with one cotyledon, narrow leaves with parallel venation.
- They are mostly herbaceous (weak stem plants.)
- Example: White, maize, grasses, tef, enset, rice, etc.

ii) **Dicotyledons** – have seeds that contain two cotyledons, broad leaves with netted venation.
- They are herbaceous or woody. Examples: Peas, sunflower, roses, beans, etc.

Comparison of Monocots and dicots

Monocots	**Dicots**
➢ The embryo has one cotyledon. ➢ Leaves have parallel – Venation.	➢ The embryo has two cotyledons. ➢ The leaves have netted venation.
➢ Steam contains vascular bundles in scattered from. ➢ Have Fibrous or adventitious roots.	➢ The vascular bundles are in ring. ➢ Have tap root system.
 Fibrous root  Adventitious root	 Taproot
➢ Have needle shaped leaves ➢ Floral parts are in 3's or multiple of three.  ➢ Have true secondary growth, with vascular cambium is absent.	➢ Have broad leaves ➢ Floral parts are arranged in 4's or 5's or multiples of 4 and 5. ➢ Have true secondary growth, with vascular cambium, present.
Example: Teff, wheat, maize	*Example*: Bean, peanut

Illustrative Example

24. The most dominant plants on earth at present are:
A. Lycophytes
B. Angiosperms
C. bryophytes
D. gymnosperms

Explanation:
Angiosperms are the most dominant plants on the earth's surface because they grow in different forms; trees, shrubs and herbs.

Answer: B

25. Plants which have true roots, leaves, stems and seeds inside the fruit are:
A. Gymnosperms
B. Angiosperms
C. Mosses
D. ferns

Explanation:
Angiosperms are higher plants that posses true roots, leaves, stems and seeds enclosed in the fruit. This is because they have flowers which consist of ovule enclosed in the ovary. The ovule develops in to a seed and ovary in to a fruit.

Answer: B

Examples No. 26 is based on the use of the following groups:
1. Sphenophytes
2. Pteridiophytes
3. Bryophytes
4. lycophytes

26. The numbers 1,2,3,4 are correctly matched with their examples as:
A. Horsetail, Ferns, liverworts, club mosses.
B. Horsetail, conifers, mosses, liverworts.
C. Orange, juniper, Ferns, club mosses.
D. Liverworts, Tid, Zigba, Fern.

Explanation: Sphenophytes (horsetails), pteridiophytes (Ferns), Bryophytes (mosses and liverworts), Lycophytes (club mosses).

Answer: A

Questions 27 and 28 are based on the following groups of organisms.
1. Arthropods
2. Angiosperms
3. Birds
4. Gymnosperms
5. Algae
6. Amphibians
7. Fish
8. Mollusks
9. Ferns
10. Mosses

27. Which of the above groups belong to the plant kingdom?
A. 1,2,3,4
B. 2,4,9,10
C. 1,3,5,8
D. 2,4,8,9

Explanation: Arthropods, birds, amphibians, fish and mollusks belong to kingdom animalia, while Angiosperms, Gymnosperms, Ferns and mosses belong to kingdom plantae. **Answer: B**

28. The branch of biology that studies about the groups listed under numbers, 1,3,6,7 and 8 is:
   A. Botany
   B. Anthropology
   C. Zoology
   D. Physiology

Explanation: **Zoology** is a branch of biology which studies about animals.

Botany is a branch of biology which studies about plants;

Answer: C

29. Which major group of the plant kingdom is best characterized by having needles shaped leaves and production of seeds in structures called cones?
   A. Ferns
   B. Gymnosperms
   C. Monocotyledons
   D. Angiosperms

Explanation: Gymnosperms are higher plants which have no flower. Their seed are not enclosed in the fruit rather their seeds are enclosed in cones. **Answer: B**

30. Which of the following alternative contains the correct characteristics of the division of plants called pteridophyta? (EGSECE 2004 E.C)
   A. No true stem, leaves and roots, vascular spore producing
   B. True stem, leaves and roots, vascular, cone producing
   C. True stem, leaves and roots, vascular, spore producing
   D. True leaves and stem, no true roots, non – vascular, flowering

Explanation: Pteridophytes are Ferns which have the following major characteristics.

- Their leaves are called **Fronds** and have underground stem called **rhizome.**
- They have **vascular tissues**. (Xylem and phloem).
- They reproduce by alternation of generation which includes sporophyte and gametophyte generations.

Answer: C

31. The organism *Rosa abyssinica* is a member of the genus:
   A. Abyssinica
   B. Plantae
   C. Rosa
   D. Angiospermea

Explanation: In scientific name the first name implies a genus name and the second name is a species name. In *Rosa abyssinica*.
Rosa is a **genus** name and ***abyssinica*** is a **species** name

Answer: C

32. Which of the following taxonomic groups contains closely related organisms?
A. Genus
B. Order
C. Phylum
D. Class

Explanation:
Arrangement of categorization according to increasing complexity is:
Species →Genus → Family → Order → Class → Phylum → kingdom, as the complexity increases the relatedness of organisms decreases.

Answer: A

33. This item is based on the following structures of plants. (EGSECE, 2005 E.C)
1. Seed 2. Flower 3. Vascular tissue 4. Cone
The structures that perfectly fit in to gymnospermae are:
A. 1,3,4
B. 1,2,3
C. 1,2,4
D. 2,3,4

Explanation: Gymnosperms are non – Flowering plants which possesses the following Features.
- Have well developed stem, root, leaf.
- Have necked seed
- Do not have flower
- Have cone which contains their seeds.

Answer: A

34. List out the common features of all plants.
Explanation:
All plants have the following common features:
- They are eukaryotes, multicellular and autotrophic
- They are cellulose cell wall and chlorophyll.
- They reproduce either sexually or a sexually.

34. Distinguish between
i) Gymnosperms and angiosperms
ii) Monocots and dicots
Explanation:
i) Comparison between gymnosperms and angiosperms.

Gymno
- They bea
- Their se
- and not e fruit.
- Their structure cones

ii) Comp

Mo
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- Have narro parallel ven
- Have fibrou roots.
- Have floral
- Vascular ca

Part I: Choose th

1. The simple
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2. Monocots
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A. Ste

Gymnosperms	Angiosperms
• They bear no flowers. • Their seeds are naked and not enclosed in the fruit. • Their reproductive structures are found in cones	• They bear flowers. • Their seeds are enclosed in the fruit. • Their reproductive organs are flowers.

ii) **Comparison between monocot and dicot**

Monocot	Dicot
• Have one cotyledon • Have narrow leaves with parallel venation. • Have fibrous or adventitious roots. • Have floral parts in 3's. • Vascular cambium is absent	• Have two cotyledons. • Have broad leaves with netted venation. • Have tap root system. • Have floral parts in 4's or 5's. • Vascular cambium present

Post Test 3

Part I: **Choose the best answer from the given alternatives.**

1. The simplest land plants are classified under division:
   A. Spenophyta C. Bryophyta
   B. Lycophyta D. Gymnospermae
2. Monocots differ from dicots by all of the following except:
   A. Have one cotyledon
   B. Have narrow leaves with parallel venation
   C. Have fibrous or adventitious roots
   D. Vascular cambium present
3. Gymnosperms do not differ from angiosperms by:
   A. They bear no flowers.
   B. Their seeds are naked and not enclosed in the fruit.
   C. Their reproductive structures are found in cones
   D. They use sexual mode of reproduction
4. Which one of the following structure is unique to angiosperms?
   A. Stem B. Root C. Flower D. Leaf

5. All are the features of kingdom plantae except:
   A. They have eukaryotic cells.
   B. They are multicellular organisms.
   C. They contain chlorophyll and carry out photosynthesis.
   D. They are heterotrophic.

6. **Kingdom Animalia**

Kingdom animalia consists of multicellular, eukaryotic, heterotrophic organisms.

- Their body has complex organ system.
- They have no chloroplasts and cell wall.
- They Feed by ingesting the food material.
- They reproduce sexually and some reproduce asexually.
- Organisms of kingdom animalia exhibit locomotion.
- They have nerve systems and sensitive to their surroundings.
- Kingdom Animalia consists of 33 phyla. Among these the major ones are:
  - Phylum porifera (sponges)
  - Phylum coelenterata (cnidaria)
  - Phylum platy helmithes (Flatworms)
  - Phylum Nematoda (Round worms and Nemathelminthes)
  - Phylum Annedida (Segmented worms)
  - Phylum Mollusca (soft – bodied animals)
  - Phylum Echinodermata (spiny – skinned animals)
  - Phylum Arthropoda (Joint – footed animals)
  - Phylunm chordata
- Animals are grouped in to two groups. These are **vertebrates** and **invertebrates.**
  i) **Vertebrates** are animals with back bone. These include chordates.
  ii) **Invertebrates** are animals without back bone. Except phylum chordata, all animal phylas are invertebrates.

Phylum Porifera (sponge)

Poriferas are the simplest of all invertebrates.

- They are multicellular animals that have a central body cavity and supportive structures called **spicules.**

- They have little coordination and control.
- They have no nervous system.
- They have hallow filter feeders.
- Only young can move by flagellar structures and Feed by engulfing food.
- Most of them are **hermaphroditic**.
- They are evolutionarily dead end and have no close living relatives.

Phlum Coelenterata (Cnidaria)

- Their bodies are composed of two layers. Namely, **ectoderm** (outer layer) and **endoderm** (inner layer).
- They are beautiful creatures and some are poisonous.
- They have tentacles with stinging organelles called **namatocysts** to poison and immobilize prey.
- They have soft body.
- They have only one opening which is mouth.
- They are **radialy symmetrical.**

NB. **Radial symmetry** is a body that can be divided into many halves which are mirror images of one on other.

Phylum Nematoda (Round Worms)

- Have unsegmented bodies and organ systems.
- They have cylindrical body.
- They have body covered with cuticles which helps them to resist drought and other unfavorable conditions.
- They have complete digestive system.
- Have **bilaterally symmetrical** bodies.
- Have no circulatory system.
- They are very important for healthy soil.
- They are found almost everywhere.
- They are the most numerous animals.

Examples: Filaria, Ascaris

Note: ***Bilateral symmetry*** is a body that can be divided in to two halves that are mirror images of one another.